# Forecast Combination with Random subsets: Accuracy Gains and Interpretability

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July 11, 2026

**Abstract**

This paper studies Random subset Regressions (RSM) as a forecast-combination strategy. We develop a finite-sample framework that makes explicit the trade-off between using more forecasts (lower combination variance) and estimating more weights (higher estimation error). Monte Carlo experiments, where individual forecasts are themselves estimated, show that RSM typically delivers lower RMSFE than equal weighting and restricted OLS. In an empirical application using real-time FRED-MD vintages to nowcast U.S. GDP growth, RSM significantly improves the forecast combination performance relative to the mean with the largest gains during the COVID-19 recession and compares favorably to Lasso, Ridge, and Random Forest benchmarks. Finally, we link RSM to Shapley value decomposition, yielding a transparent decomposition of the combined forecast into contributions of individual forecasts even in high-dimensional settings.

**Keywords:** Forecasting, Forecast combination, Random Subset, Forecast Combination Puzzle.

**JEL:** C53, C52, C32.

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# 1 Introduction

Nowcasting, as well as short-horizon forecasting of economic activity, is crucial for decision-makers such as firms and governments. Over the last couple of decades, the amount of data used for (improving) macroeconomic forecasting has been increasing sharply. This means researchers may generate hundreds of different forecasts for a given macroeconomic indicator, for example, GDP. These individual predictions may be derived from diverse methods or data sources. A natural question arises: how to combine and/or select from numerous forecasts in order to produce a final prediction?

One of the most straightforward techniques used by practitioners and researchers is to assign equal weight to each prediction. This simple aggregation method proves to be a solid benchmark and often outperforms more complex combination rules. In the literature, this fact is known as a "forecast combination puzzle". However, despite its straightforward implementation and often empirical robustness, in many cases, the rule of applying equal weight to each individual forecast can be harmful. The theoretical advantage of using the ad hoc equal-weighting is that it avoids estimating weights and therefore carries no estimation error. The latter can be substantial if the number of forecasts is rather high (especially compared to the time dimension). Still, in many practically-relevant cases, taking the simple average is rather restrictive since it does not take into account any prior historical performance of predictions. Equal weighting cannot detect genuine heterogeneity in forecast quality. Intuitively, by unconditionally taking the mean of a set of forecasts, one is likely to misspecify the optimal weights: too much weight to weak forecasters and too little to strong ones. This can be especially important during turbulent times such as the COVID-19 crisis, when some of the forecasts (for instance, if they are built using the leading indicators) respond substantially differently to the shock.

On the opposite side,

Economists and policymakers care not only about predictive accuracy, but also about understanding the forces driving the forecasted outcomes. Forecasting (and forecast combination) should help motivate economic and public policy decisions rather than deliver intractable, though sometimes accurate, predictions (Burgess et al., 2013). Even the "best" single model can occasionally produce large errors. In principle, understanding a model's mechanics makes it easier to diagnose unusual predictions in light of the inputs and the underlying economic environment.

In this article, we propose a framework that connects two seemingly independent fields: forecast combination and model interpretability. We build on the Random subset Regression method (RSM) Boot and Nibbering (2019), originally proposed for forecasting rather than forecast combination. A specific advantage of RSM is that this model makes it feasible to remain within a linear framework even when the number of predictors (here, individual forecasts) is large. Hence, it also allows for obtaining the contributions for each variable directly via developed shortcuts. Essentially, RSM estimates many low-dimensional regressions on randomly selected subsets of forecasts and then aggregates the resulting subset predictions. Thus, it replaces a single high-dimensional (and potentially infeasible) regression with a collection of smaller, related estimation problems, therefore reducing weight-estimation error while preserving or even improving forecast performance.

First, we study the theoretical properties of RSM for forecast combination. We show that in population RSM unifies both the equal weighting and the optimal linear combination under a sum-to-one restriction (restricted OLS). In other words, both benchmarks are a special case of RSM and the latter interpolates between the two; therefore, being more flexible in practice. Furthermore, in the exogenous common-loading case, we develop a finite-sample framework that helps to analyze the trade-off between using more forecasts, which reduces approximation error, and estimating more combination weights, that in turn increases finite-sample estimation error. The results show that the excess risk of RSM can be defined as the covariance-weighted distance between the averaged random-subset weight vector and the oracle weight vector. This representation makes transparent why RSM can improve on equal weighting when forecasts are heterogenous enough. Simulatenously, it the conditions under which RSM achieves a lower Mean Squared Error (MSE) than both and the equal weighting benchmark. The results are in line with, for example, Elliott and Liao (2026) in the sense that in the exogenous case – when forecasts are treated as given – in order to beat the equal weighting scheme, RSM requires overcoming the estimation penalty. Simultaneously, RSM can mitigate this penalty relative to estimating the full set of optimal weights. Also, the general expressions as well as closed-form ones (in some specific cases) are obtained to illustrate the subset size $k^*$, which minimizes MSE under the RSM framework even if in a specific case (such as compound symmetry) RSM cannot outperform equal weighting because the latter is itself optimal. The analysis implies, that $k^*$ depends strongly but not exclusively on the sample size $T$. In particular, we show that $k^* \propto \sqrt{T-1}$ and often takes a form $k^* = c\sqrt{T-1}$, where $c$ is some constant which can depend on several other model-specific values, for instance, the pairwise correlation among forecasts.

Second, we conduct the Monte Carlo simulations to evaluate the RSM forecast combination performance compared to the same benchmarks (equal weighting and restricted OLS "optimal" weights). Unlike the theoretical analysis, in this exercise, we consider a different setting. Namely, when the forecasts are not exogenous, but rather generated from each variable included in the data-generated process (DGP) of the dependent variable. This design – univariately constructing covariate-based predictions – is standard in the forecast combination literature when a high-dimensional setting is considered (among others, see Chen and Maung (2023), Samuels and Sekkel (2017)). Moreover, this specification helps us to understand how the theoretical results derived for the exogenous case translate to cases when the forecasts themselves are estimated.

The Monte Carlo simulations results suggest that RSM yields the lower root mean squared forecasting error (RMSFE) than both benchmarks when the subset dimension is chosen near the $k^*$ across all DGPs considered. However, the size of the gains depends on various factors, including the time dimension $T$. Consistent with the provided theoretical results, equal weighting is often more resilient than the optimal weights in the case when the number of forecasts is large, because the latter suffers from the weight-estimation error.

Typically, RMSFE, as a function of $k$, takes a $U$-shaped form highlighting a non-linear relationship. This finding indicates the importance of choosing the right $k$: if the subset size is too high or too low, the relative performance of RSM can deteriorate. However, the value of $k^*$ which minimizes RMFSE across all scenarios tends to lay in the close proximity to $k = \sqrt{T}$ rounded to the nearest integer. In most cases, there are no additional gains of estimating true $k^*$ via cross validation. The ad hoc rule of choosing $k = \sqrt{T}$ yield the similar to (sometimes exact) minimum RMSFE value. Thus, this choice of $k$ is an empirically robust rule of thumb for the endogenous simulation and empirical settings considered. Therefore, we treat $k = \sqrt{T}$ as a practically relevant starting point for applying RSM for forecast combination in the endogenous case. Alternatively, the cross validation, if desired, can be implemented around the proposed value.

Additionally, we explore a synergy between RSM and the SVD. Because RSM implies a linear structure at the subset level, we can decompose the RSM prediction into forecast contributions using standard linear-model shortcuts, while still accommodating high-dimensional sets of forecasts. Thus, under this framework, it is possible to circumvent the challenges of using many variables while dealing with the SVD. We make sure that the main SVD advantage is fulfilled: the difference between the final prediction and a baseline (e.g., the historical mean) is fairly distributed among the individual forecasts. That is, all contributions sum up exactly to the total deviation holding up to an intercept term.

In our empirical application, we use real-time vintages of the FRED-md dataset, provided by the Federal Reserve Bank of St.Louis, to nowcast US GDP using a high-dimensional set of predictions. For the main analysis, the optimal subset size is set to $k = \sqrt{T}$ rounded to the nearest integer, motivated by the Monte Carlo simulation results. We also examine robustness to alternative values of $k$ and found only marginal gains of using $k^*$, found by cross-validation. The results indicate that RSM outperforms the equal weighting, delivering lower RMSFE across the specifications. In the full sample, the relative to the mean RMSE for RSM is 0.67 and 0.88, if the COVID-19 quarters are excluded. This strengthens the intuition that during crises, the RSM produces a more accurate forecast combination due to its flexibility. Additionally, we also compare RSM to other high-dimensional benchmarks such as LASSO, Ridge regression, and Random Forests, with RSM achieving the lowest RMSE among the competing approaches. These empirical evidence supports the practical usefulness of the rule-of-thumb choice of $k = \sqrt{T}$, even though estimating $k^*$ can sometimes yield further improvements.

Finally, we empirically investigate the difference between the weights and corresponding Shapley Values obtained by the equal weighting and RSM, respectively. We observe that equal weighting tends to underestimate the global weight of some important predictions (for example, industrial production), which, in turn, leads to underestimating the relative contribution of this forecast, especially during the COVID-19 crisis. The tight link between weights and contributions also potentially helps interpret negative weights, which are often puzzling in practice.

**Literature review and contributions.** This paper relates to several strands of literature. First, this paper contributes to the literature of forecast combination by providing a flexible and easy-to-implement regression-based estimation of forecast combination weights using random subsets. So far, RSM and its various modifications is used mostly to produce forecasts based on the actual data (Boot and Nibbering (2019), Wichitaksorn et al. (2023), Kotchoni et al. (2019), Pick and Carpay (2022), Kant et al. (2025), Meligkotsidou et al. (2021), Dinh et al. (2024)) with little being known about its properties in the forecast combination framework. One closely related contribution is Chen and Maung (2023), who, in the spirit of the complete subset regression idea of Elliott et al. (2015), fix a subset size $k$, estimate many $k$-variate forecasting regressions, and then average the resulting subset-model forecasts. In their empirical implementation, $k$ is kept small (e.g., $k \in 1, 2, 3, 5$) largely because the number of candidate subsets grows exponentially with $k$, making larger values computationally burdensome. By contrast, our random-subset approach treats subsetting as an integral part of the forecast-combination estimator. We repeatedly estimate constrained combination weights within randomly drawn subsets and use a theory-guided estimation-error trade-off to motivate the choice of $k$. Our framework yields practical guidance for selecting $k$ (typically higher that proposed values used in Chen and Maung (2023)). Consistent with this perspective, Chen and Maung (2023) report that forecasting performance improves as $k$ increases over the small grid they consider, suggesting that further gains may arise.

Second, we contribute to the growing literature of high-dimensional model/variable selection and dimension reduction in the forecast combination settings. While dealing with a large number of variables, researchers often assume sparsity. Sparse models explicitly imply that the least relevant variables are discarded. For example, LASSO, as proposed in Tibshirani (1996), forces coefficients for many predictors to equal zero, which can be too restrictive. RSM avoids assuming sparsity – an assumption often imposed in the high-dimensional settings. Moreover, Giannone et al. (2017) argue, based on Bayesian forecasting exercise, that posterior predictive distributions are characterized by a combination of various different models rather than being concentrated on a single sparse model or a single dense model. This intuition can transfer to the frequentist forecast combination settings suggesting that a well-designed model averaging technique can outperform any sparse model, as shown for the forecasting case in Kotchoni et al. (2019).

Furthermore, RSM allows for computing each predictor's contribution, linear attribution scheme inspired by Shapley Value Decomposition.

The rest of the paper is organized as follows: Section 2 provides the current state of the forecast combination research. In Section 3, we describe the RSM method and how it can be used to combine forecasts. Section 4 considers the theoretical framework. Section 5 explains the Monte Carlo framework and simulation results. Section 6 introduces Shapley Values and establishes their correspondence to RSM. Section 7 contains an empirical application of the RSM method. Finally, Section 8 concludes and provides an outlook for future research.

# 2 Forecast Combination

Many researchers from various disciplines have analyzed a wide range of forecast combination methods. In many practical settings, there are multiple forecasts of the same variable obtained either by different methods or by using different sources of information (Timmermann, 2006). A common premise is that combining different individual predictions can help to improve accuracy. The aggregation strategy enables the incorporation of much more available information, which indeed may be more fruitful than trying to seek a "best" model. For a review of the extensive literature on forecast combinations, see Wang et al. (2023). In this section, we describe methods dealing with point forecast combinations based on different linear approaches. We focus on this specific class of forecast aggregation because, as can be seen further, it is possible to merge the latter with RSM. To conclude, a forecast combination puzzle is discussed.

## 2.1 Linear Combinations of Individual Forecasts

One of the most straightforward strategies to combine various forecasts is to assign weights to each individual prediction: the more precise the forecast, the greater the weight. There is a plethora of different weighting schemes. Thus, it sounds feasible to construct a linear combination of the individual forecasts and treat the OLS coefficients as weights.

More formally, suppose an $m$-dimensional vector of $h$-step ahead forecasts $\hat{\boldsymbol{y}}_{T+h|T}$ is given. Additionally, an $T$-dimensional vector of past observations of the targeted variable $\boldsymbol{y_T}$ is observed. Under this framework, regression-based weights can be defined as follows (Granger and Ramanathan, 1984):

$$y_{T+h} = w_{T+h|T,0} + \boldsymbol{w'_{T+h|T}} \hat{\boldsymbol{y}}_{T+h|T} + \epsilon_{T+h}, \tag{1}$$

where $\hat{\boldsymbol{y}}_{T+h|T}$ are past individual forecasts.

Some restrictions may still be imposed on $\boldsymbol{w'_{T+h|T}}$. For instance, a constant term can be omitted, and/or the weights can be adjusted so that their sum adds up to one. In the latter case, we obtain the so-called "optimal" weights that are obtained by running restricted OLS regression Bates and Granger (1969):

$$y_{T+h} = \boldsymbol{w'_{T+h|T}} \hat{\boldsymbol{y}}_{T+h|T} + \epsilon_{T+h}, \quad \sum \boldsymbol{w'_{T+h|T}} = 1 \tag{2}$$

The use of regression-based weights for forecast combinations has a long history. Both advantages and limitations of this approach have already been examined in the last century. One of the critiques is shown, for example, in De Menezes et al. (2000). While exploiting the unrestricted regression, one should address potential multicollinearity and serial correlation in forecast errors. We assume that both of these challenges are mitigated when combining this approach with RSM, as discussed later.

Upon establishing the regression-based weighting method, another concern arises. What if the total number of $\hat{\boldsymbol{y}}_{T+h|T}$ is exceptionally large? In this case, a simple OLS estimation becomes impractical, particularly when the number of forecasts (N) exceeds the number of observations in the training set (n). It makes sense to use some variable selection methods, such as Least Absolute Shrinkage and Selection Operator (LASSO) (Tibshirani, 1996), to end up only with a handful of individual forecasts.

Typically, when dealing with high-dimensional data (in the form of $N > n$), LASSO or other penalized regressions have been extensively used for dimension reduction in many areas, including economics. However, sometimes it is necessary to use as much data as possible without excluding variables by assuming non-sparsity. RSM can manage this level of selection 'strictness' by utilizing all the available data.

### 2.2 Forecast Combination Puzzle

Although various forecast combination methods have been proposed over the last couple of decades, in practice, the simple average (assigning equal weights to each of the forecasts) tends to perform similarly to (or even better than) more sophisticated aggregation approaches. This fact perplexed researchers, who had been trying to find a prolific answer to that paradox. Stock and Watson (2004) were among those who studied this phenomenon. In that paper, the term "Forecast combination puzzle" was introduced.

Indeed, one should expect some gains in terms of forecast accuracy while using more rigorous tools for forecast combination rather than averaging. Moreover, theoretical results for some approaches were formally derived, proving this initial claim. However, often a simple mean overshadows other weighting schemes. Again, in Wang et al. (2023), a great summary of why the "Forecast combination puzzle" holds empirically is presented. To summarize, a possible explanation is that the weights are misspecified or wrongly computed due to, for instance, structural changes or limited data availability. Alternatively, benefits from using a sophisticated combination may be relatively small, thus estimation per se overwhelms the potential gains, especially if the weights are treated as random and not fixed Claeskens et al. (2016).

However, if individual forecasts come from different sources or models, the variance can be high, in particular due to some outliers. If the number of outliers is high and they are not equally distributed around the mean, then averaging over this forecasting set may not be efficient. Although taking a median can arise as a potential method to produce a final prediction, as used in Stock and Watson (1999), it does not indicate how varied the individual predictions are in the set. Moreover, individual forecasts, even without outliers, can be distributed in such a way that taking a mean can be almost uninformative. For example, in Figure 1 and in Figure 2, the histograms for the expected change in prices during the next year based on the Surveys of Consumers, provided by the University of Michigan, for November 2024 and July 2025 are presented.

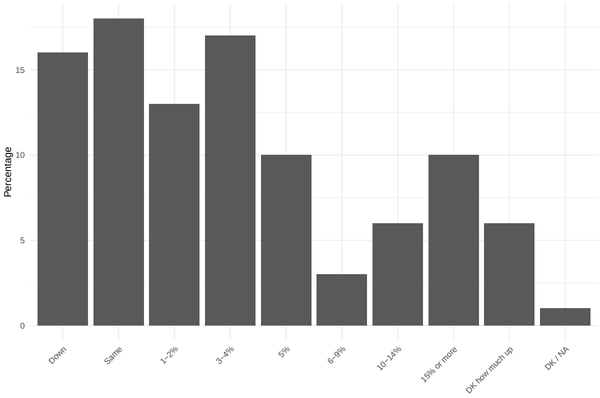


Figure 1: Expected Change In Prices During the Next Year (November 2024)

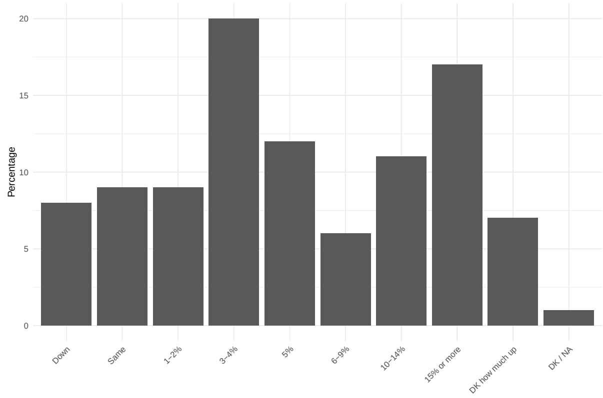


Figure 2: Expected Change In Prices During the Next Year (July 2025)

Because the Survey of Consumers extracts inflation expectations in pre-defined categories rather than as continuous numerical responses, any "mean expectation" is not directly observed but constructed by assigning representative values to bins. This construction is particularly consequential in the presence of open-ended categories such as "15% or more" and "Down", where the set of admissible values is wide, and the bin midpoint is undefined. Both histograms exhibit fat right tails, so the mean (as reported in the survey) can be disproportionately influenced by a minority of extreme responses and may therefore provide a poor summary of the "typical" respondent's beliefs.

For these reasons, aggregation methods that are less tail-sensitive are often more informative than a single mean. Using RSM for forecast combination allows for relying on the mean even in the presence of outliers and/or if there are fat tails in the initial set. This is achieved by obtaining a new set of forecasts – based on the initial one – which is supposed to be distributed in a way that taking a mean makes sense and provides good empirical performance.

## 3 Random subset

Initially, RSM methods appeared in the field of machine learning without much application in macroeconomic forecasting. Ho (1998) introduced this approach to train tree models, later Bay (1998) extended the same logic to the nearest neighbour classifiers. Wichitaksorn et al. (2023) applied RSM to feature selection in a logistic regression framework. Recently, random or complete[1] subset methods have become a powerful tool used in macroeconomics (Elliott et al. (2013),Kotchoni et al. (2019),

[1] Complete subset methods imply that for a given set of predictors, forecasts from all possible linear regressions are combined.

Pick and Carpay (2022)) and finance (Meligkotsidou et al. (2021)). Some further examples of papers utilizing subset methods can be seen in **?**.

Random subset regression is a randomized reduction method, according to which a small number of indicators are (randomly) selected to estimate a lower-dimensional problem. This procedure is repeated a sufficient number of times. Thus, one ends up with many approximations to the initial – sometimes infeasible to calculate directly – model. Finally, the forecasts from each low-dimensional model are combined (averaged). This allows for reducing forecast variance while utilizing all available information.

More formally, suppose that the following data-generating process (DGP) is defined as follows:

$$y_{t+1} = x_t'\beta + \epsilon_{t+1}, \tag{3}$$

for $t = 1, \cdots, T$, $x_t'$ is a vector of predictors in $\mathbb{R}^N$, $\epsilon_{t+1} \sim iid(0, \sigma^2)$.

Initially, this model has been proposed for using observed time series to predict the target. Instead, we use this approach for forecast combination. To draw a parallel with previous sections and align the notation, in Equation 3, $x_t'$ corresponds to the $N$-dimensional set of individual forecasts, whereas the $n$-dimensional vector $y_{t+1}$ is the final prediction of the independent variable.

As discussed previously, when the number of estimated coefficients is large, it is either impossible to run OLS or there is a high variance of the latter, which can significantly decrease forecast accuracy. To overcome both potential challenges, following Boot and Nibbering (2019), it is possible to make a projection of $N$-dimensional vector $x_t$ onto a $k$-dimensional subset using a permutation matrix, denoted $R_i \in \mathbb{R}^{N \times k}$, such that:

$$\tilde{x}_t' = x_t' R_i$$

A permutation matrix $R_i$ is responsible for randomly selecting (without replacement) a subset of $k$ predictors out of all $N$ covariates. To illustrate this idea, imagine a simple example where there are $N = 5$ variables and $k = 3$ of those are drawn. One possible $R_i$ can look like as follows (**?**):

$$R_i = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}$$

It should be noted that there are different methods to generate $R_i$. Instead of selecting a subset of covariates, weighted averages can be used to construct a different set of predictions. Weights can be chosen randomly from a normal distribution (random projections). Moreover, the proposed specification of RSM implies that the selection is carried out using a uniform distribution (the probability that a given variable is selected is equal across the covariate set). It is also feasible to use so-called importance sampling probabilities or leverage scores (Mahoney et al., 2011). The rest of the paper uses coordinate subsampling via a 0–1 permutation matrix with uniform inclusion probabilities via RSM.

Assigning an equal probability to each of the covariates can be partially explained by the following logic. Suppose there is a concern that, for a given model, randomly selected covariates might be less relevant than the rest of the pool. At the same time, most (if not all) of the time series used for macroeconomic predictions are correlated. That fact hints at two possible conclusions. First, a "less" relevant covariate is likely correlated with one that possesses more predictive power. Moreover, repeating this procedure – drawing variables randomly and averaging results over each iteration – makes the model less prone to 'uninformative' variable selection.

Conditional on a drawn $R_i$, a $k$-dimensional vector of OLS coefficients $\hat{\beta}^i$ can be obtained, as well as a single aggregated forecast value of $\hat{y}_{t+1}$. We are interested in both statistics. First, by averaging different $\hat{y}_{t+1}$, we claim that the RSM method could be beneficial for forecast aggregation, providing better predictions compared to more standard metrics. Later, we demonstrate the forecast performance of this method compared to benchmarks.

## 4 Theoretical Model

Suppose, there are $m$ individual forecasts $f_t = (f_{1t}, \ldots, f_{mt})'$ of a scalar target variable $y_t = \mu_t + \epsilon_t$. Each forecast in $f_t$ is assumed to be unbiased. Additionally, the vector of forecast errors relative to the conditional target component is to be defined. The $m$-dimensional vector of forecasts errors is denoted as $u_t = f_t - \mu_t \iota_m$, where $\iota_m$ is the $m$-dimensional vector of ones. Let $\mathcal{F}_\sqcup$ be an information set which is a a filtration generated by the past values of $y_t$ and potentially some unobserved covariates that can be used to construct forecasts $f_t$.

**Assumption 1.** *Target innovation:* $\mathbb{E}(\epsilon_t) = 0$, $Var(\epsilon_t) = \sigma_\epsilon^2 > 0$, $\mathbb{E}(\epsilon_t u_{it}) = 0 \quad \forall i$.

**Assumption 2.** *Forecast-error covariance:* $\mathbb{E}(u_t) = 0$, $Var(u_t) = \Sigma_{e,m} \succ 0$.

Following Elliott and Liao (2026), we can rewrite the setup as follows[2]:

$$E\left[\begin{pmatrix} y_t \\ f_t \end{pmatrix} \middle| \mathcal{F}_{t-1}\right] = \begin{pmatrix} \mu_t \\ \mu_t \iota_m \end{pmatrix}, \qquad \mathrm{Var}\left[\begin{pmatrix} y_t \\ f_t \end{pmatrix} \middle| \mathcal{F}_{t-1}\right] = \begin{pmatrix} \sigma_\epsilon^2 & 0 \\ 0 & \Sigma_{e,m} \end{pmatrix}.$$

We treat the set of $m$ candidate forecasts as exogenous draws from an unbiased forecast distribution conditional on information at time $t$. $\Sigma_{e,m}$ summarizes cross-forecast dispersion: its diagonal elements capture the marginal forecast-error variance (no forecasters know the true model) and its off-diagonals capture how strongly forecast errors co-move.

[2]Without loss of generality, we consider a fixed forecast horizon $t + 1$. The latter subscript is omitted for simplicity.

This setup allows for constructing the Mean Squared Error (MSE) loss function of any combined forecast if the latter is obtained by a weighted average (linear combination) of given forecasts under the assumption that the weights sum up to one. That is, if one produces a point forecast $w'_m f_{it} - y_t = w' u_t - \epsilon_t$ such that $\sum \omega'_m = 1$, the desired loss function has the following form[3]:

$$L = \underbrace{(1 - \omega'_m \iota_m)^2 \mu_t^2}_{Bias} + \underbrace{\sigma_\epsilon^2 + \omega'_m \Sigma_{e,m} \omega_m}_{Variance} \tag{4}$$

However, when combining unbiased forecasts under the constraint that their corresponding weights add up to one, the final prediction will be unbiased. Thus, the first (bias) term in the Equation 4 is zero, and the comparison across such different methods comes down to minimizing the variance estimation part, which is equal to $\sigma_\epsilon^2 + \omega'_m \Sigma_{e,m} \omega_m$.

The loss function depends on two entries: the vector of weights $\omega_m$ and the variance-covariance matrix of individual forecasts $\Sigma_{e,m}$. Notice the target innovation $\sigma_\epsilon^2$ is common to all forecasts combinations and refers to the irreducible noise shared by all combinations.

For both equal weighting and optimal weights, it is possible to derive the closed-form expression for the loss function. When averaging, each weight is the same for each forecast. Therefore, the vector of weights and the loss, in this case, is equal to:

$$w^{ave} = m^{-1} \iota_m, \quad L^{ave} = \sigma_\epsilon^2 + m^{-2} \iota'_m \Sigma_{e,m} \iota_m. \tag{5}$$

The weights and corresponding loss for the optimal combination can be denoted as

$$w^{opt} = (\iota'_m \Sigma_{e,m}^{-1} \iota_m)^{-1} \iota'_m \Sigma_{e,m}^{-1}, \quad L^{opt} = \sigma_\epsilon^2 + (\iota'_m \Sigma_{e,m}^{-1} \iota_m)^{-1}. \tag{6}$$

Finally, a specific estimation error form of the loss $L^{opt}$ is defined. This inflation factor, denoted $\mathcal{I}(T,k)$, is due to the estimation error arising from running restricted OLS. There are several potential choices of the its form. For example, Elliott and Liao (2026) assume so-called "best case scenario", where it is characterized by the expression $\mathcal{I}(T,m) = (1 + \frac{m-1}{T})$. This multiplicative term is a reduced-form way to capture finite-sample estimation noise from estimating $m-1$ free weights under the sum-to-one constraint. Instead, we adopt a reduced-form approximation to the out-of-sample inflation generated by estimating $k-1$ free combination weights. In this case, this factor is $\mathcal{I}(T,m) = \frac{T-1}{T-m}$. It is possible to see that Elliott and Liao (2026) use the first order approximation of this factor which yields under $m << T$:

$$\frac{T-1}{T-m} = 1 + \frac{m-1}{T-m} \approx 1 + \frac{m-1}{T}$$

[3]This holds if estimated weights are evaluated on a new observation independent of the training sample. Conditional on the training sample, estimated weights are fixed.

Therefore, the loss for the optimal combination weights could be approximated by the following expression:

$$L^{opt} \approx (\sigma_{\varepsilon}^2 + Q^{opt}) \left( \frac{T-1}{T-m} \right), \tag{7}$$

where $Q^{opt} = (\iota_m' \Sigma_{e,m}^{-1} \iota_m)^{-1}$.

This expression of the loss function is used later to compare optimal weights to the weights obtained by RSM[4]. Notice, there is no estimation penalty for the equal weighting since no weights are estimated. Thus, the benchmark loss for equal weighting is as follows:

$$L^{ave} = \sigma_{\epsilon}^2 + Q^{ave},$$

where $Q^{ave} = m^{-2} \iota_m' \Sigma_{e,m} \iota_m$.

This setup allows to show the importance of the irreducible noise $\sigma_{\epsilon}^2$. Boot and Nibbering (2019), inter alia, subtract this term prior to further calculations since a standard assumption is that it plays no role in choosing $\omega_m$. However, as shown in Elliott and Liao (2026), the irreducible component goes directly into loss function which is inflated by the estimation error. That is why, ignoring the irreducible noise can potentially alter further arguments. We adopt the framework used in Elliott and Liao (2026), thus $\sigma_{\epsilon}^2$ enters our model.

## 4.1 Random subset Method for Forecast Combination

We start from the simple example when one random subset $S \subset \{1, \dots, m\}$ of size $k$ is drawn uniformly without replacement from $\binom{m}{k}$ possible subsets. Let $\Sigma_{e,k,S}$ denote the $k \times k$ principal submatrix of $\Sigma_{e,m}$ indexed by $S$. Inside subset $S$, the population optimal subset weights are as follows:

$$\omega_S = \frac{\Sigma_{e,k,S}^{-1} \iota_k}{\iota_k' \Sigma_{e,k,S}^{-1} \iota_k}. \tag{8}$$

We embed these weights into $\mathbb{R}^m$ by defining $v_S = P_S w_S$, where $P_S$ places the $k$ selected weights in their original coordinates and sets all other $m-k$ coordinates to zero. Then, $v_S' \iota_m = 1$, which ensures that the weights still sum up to one. More formally, we can write:

$$v_S = P_s \omega_S = P_s \frac{\Sigma_{e,k,S}^{-1} \iota_k}{\iota_k' \Sigma_{e,k,S}^{-1} \iota_k}, \quad \sum_{i=1}^{m} v_S = 1. \tag{9}$$

Introducing $v_S$ allows us to define three separate population risks. This can be achieved because the matrix $P_S$ zero-pads the weights from subset $S$ of size $k$ into $\mathbb{R}^m$. Therefore, the embedded full-dimensional form and its subset counterpart are the same:

[4]For RSM, subset estimates overlap in regressors, so their estimates are correlated. Therefore, the inflation factor is an approximation inspired by Elliott and Liao (2026).

$$v_S'\Sigma_{e,m}v_S = \omega_S'\Sigma_{e,k,S}\omega_S$$

First, the average subset population risk that uses a single draw $S$ and computes the expected loss over all possible subsets.

**Definition 1** (Average subset RSM)**.** *The average-subset population risk is defined as:*

$$Q_k^{\text{sub}} = \mathbb{E}_S[v_S'\Sigma_{e,m}v_S] = \mathbb{E}_S[Q(S)], \tag{10}$$

*where*

$$Q(S) = (\iota_k'\Sigma_{e,k,S}^{-1}\iota_k)^{-1}. \tag{11}$$

The average subset population risk takes a random subset draw of $k$ forecasts, computes the population-optimal weights within the chosen subset $v_S$, and reports the single forecast $v_S'f_t$. On the population level, it computes the expected loss over the random draw of $S$. In other words, it averages the subset risk across all possible subsets before a single one is drawn. This means, intuitively, a decision-maker who chooses one draw gets a risk oscillating around $Q_k^{sub}$ depending on which $S$ is drawn at random from all $\binom{m}{k}$ subsets of size $k$.

Alternatively, $Q_k^{sub}$ can be viewed as the average of losses across subsets. However, RSM, as described in this paper, follows a different procedure. It first averages the weight vectors – and forecasts themselves – then evaluates a single loss value based on averaging. To describe the latter approach, we can define two versions of the potential losses that depend on the number of draws.

First, we consider a case when a finite amount of draws are taken.

**Definition 2** (Finite-bagged RSM)**.** *Suppose, $G$ independent subsets $S_1, \ldots, S_G$ of size $k$ are drawn uniformly at random. The averaged weight vectors in this case can be represented as:*

$$\hat{v}_{k,G} = \frac{1}{G}\sum_{g=1}^{G} v_{S_g}. \tag{12}$$

*Thus, the final forecast is $\hat{v}_{k,G}'f_t$. Unlike the average subset case, the loss is computed for the aggregated forecast rather by averaging individual losses per subset. The finite-bagged population risk, averaging over subset draws, is as follows:*

$$Q_k^{\text{bag},G} = \mathbb{E}[\hat{v}_{k,G}'\Sigma_{e,m}\hat{v}_{k,G}].$$

$Q_k^{\text{bag},G}$ characterizes the loss after finitely many selections of subsets. On practice a decision-maker gets this value as potential loss after using RSM. Therefore, this quantity is crucial in the analysis. However, it is possible to explore the case when $G \to \infty$, thus obtaining an infinite-bagged RSM.

**Definition 3** (Infinite-bagged RSM)**.** *Suppose you take infinitely many draws of subsets of size $k$. Then, the infinite-bagged RSM weight vector is defined as:*

$$\bar{v}_k = \mathbb{E}_S[v_S]. \tag{13}$$

*Therefore, the infinite-bagged population risk can be expressed as follows:*

$$Q_k^{\text{bag},\infty} = \bar{v}_k' \Sigma_{e,m} \bar{v}_k. \tag{14}$$

Notice that $Q_k^{\text{bag},\infty}$ describes the deterministic risk of the fixed weight vector $\bar{v}_k$. This is achieved with the help of the law of large numbers as $G \to \infty$. Moreover, due to convexity of the population risk function in $v_S$, we can show by applying Jensen inequality that:

$$Q_k^{sub} = \mathbb{E}_S[v_S' \Sigma_{e,m} v_S] \geq (\mathbb{E}_S[v_S])' \Sigma_{e,m} (\mathbb{E}_S[v_S]) = Q_k^{bag,\infty} \tag{15}$$

This means that infinite bagging produces the loss not greater than the single-draw estimator since the assumption $\Sigma_{e,m} \succ 0$ makes $v' \Sigma_{e,m} v$ strictly convex. The equality occurs when $v_S$ is constant across subsets. The case $v_S = \bar{v}_k$ happens only when $k = m$. Whenever $v_S$ is not constant across subsets, the Inequality 15 is strict.

After defining three population risks, it is possible to show the relationship between them:

**Proposition 1** (Finite-bagging bridge)**.** *For independent subset draws,*

$$Q_k^{\text{bag},G} = Q_k^{\text{bag},\infty} + \frac{1}{G} \left( Q_k^{\text{sub}} - Q_k^{\text{bag},\infty} \right). \tag{16}$$

*Consequently,*

$$Q_k^{\text{bag},\infty} \leq Q_k^{\text{bag},G} \leq Q_k^{\text{sub}}.$$

The proof can be found in Appendix A.

Equation 16 shows two important takeaways. First, it relates three population losses to each other regardless of distributional assumption on $\{f_t\}$. The expected finite loss can be represented as its infinite counterpart plus the bagged error – the averaged difference between the infinite expected loss and the expected loss of a single draw. Therefore, it depicts the fact that increasing $G$ monotonically reduces the finite-bagged population risk until the infinite-bagged population risk is reached. It is worth mentioning a couple of special cases:

- when $G = 1$, $Q_k^{bag,1} = Q_k^{sub}$
- when $G \to \infty$, $Q_k^{bag,G} \to Q^{\text{bag},\infty}$

The latter special case is particularly valuable. It illustrates that if the number of draws $G$ is rather large, $Q_k^{bag,G}$ can be arbitrarily close to its infinite bagged counterpart $Q_k^{bag,\infty}$.

Second, Equation 16 shows the ordering of the losses. Through Inequality 15, we show $Q_k^{sub} \geq Q_k^{bag,\infty}$. On the other hand, Equation 16 places $Q_k^{\text{bag},G}$ in-between the other losses. In other words, the finite bagged RSM is an intermediate case between the infinite bagged version and average subset RSM, which can be arbitrarily close to former if $G$ is large enough. Therefore, for the rest of the paper, we assume that $G$ is selected such that the bagged error is close to 0. Correspondingly, $Q_k^{bag,G} \approx Q_k^{bag,\infty}$.

### 4.2 Global and Conditional RSM Weights

We can analyze how the RSM weights change depending on the subset size $k$ in some general cases. First, suppose that $k = 1$. This means that every subset is a singleton $S = i$, where only one variable $i$ is selected. Within this subset the constrained OLS assigns the full weight to the drawn forecast and zero to all other forecasts. Namely:

$$v_{\{i\},i} = 1, \quad v_{\{i\},j} = 0 \quad \forall j \neq i$$

Since each singleton $\{i\}$ is drawn uniformly at random with probability $\frac{1}{m}$, the infinite-bagged weight can be represented as:

$$\bar{v}_{1,i} = \mathbb{E}_S[v_{S,i}] = \frac{1}{m} \cdot 1 + \frac{m-1}{m} \cdot 0 = \frac{1}{m} \implies \bar{v}_1 = \frac{1}{m} = w^{ave}$$

Thus, in population, the infinite-bagged RSM with $k = 1$ produces the equal weights.

Similarly, we can study the case when $k = m$. Then, there is only one full subset $S = \{1, \ldots, m\}$ which excludes the random selection. In this case, the constrained OLS produces optimal weights:

$$\bar{v}_m = v_S = (\iota_m' \Sigma_{e,m}^{-1} \iota_m)^{-1} \iota_m' \Sigma_{e,m}^{-1} = w^{opt}$$

The averaging over $G$ random subsets is not necessary in this case since all subsets are the same – full subset, to be precise. Therefore, when $k = m$, one constrained OLS will produce the same result even if you repeat it $G$ times due to unchanged subset selection.

**Proposition 2** (RSM as a unifier)**.** *Both equal weighting and optimal weighting are the special cases of RSM. If $k = 1$, RSM collapses in population to equal weighting, whereas if the full set of $m$ forecasts is used, RSM produces optimal weights.*

These special cases when $k = 1$ and $k = m$ reveal an important conclusion. It contains both these standard benchmarks inside itself. This conclusion shows the flexibility of RSM.

If forecasts are homogeneous such that their optimal weights are equal to or close to equal weighting, RSM presumably detects it through showing the lowest loss is achieved with small $k$. On the contrary, if using all available forecasts is feasible (in the case when $T >> m$), then RSM selects $k = m$. However, in many real-world applications the forecasts are heterogeneous enough to make equal weighting unreliable whereas computing full optimal weights on the short panel of forecasts when $T$ is not large compared to $m$ yields noisy estimations. This is why, the intermediate solution $1 < k < m$ can produce the lower loss in comparison to the benchmarks. This means, instead of choosing between equal weighting and optimal weighting schemes ad hoc, one may estimate RSM with different $k$ and presumably get the results not worse than any of the benchmarks.

So far, we refer only to the global RSM weight. However, it is possible to establish a correspondence between a global weight and a conditional weight. The latter is conditional on being select in a subset $S$. More formally, when subsets of size $k$ are selected uniformly at random we can write the RSM global weight by definition as follows:

$$\bar{v}_{k,i} = \mathbb{E}_S[v_{S,i}] = P(i \in S) \cdot \mathbb{E}_S[w_{S,i}|i \in S] = \frac{k}{m}\mathbb{E}_S[w_{S,i}|i \in S] \tag{17}$$

Equation 17 illustrates how the global and conditional weights are related to each other. Suppose after using RSM, a forecast $i$ has a weight greater than the equal weight. In this case:

**Proposition 3** (Global and conditional weights correspondence)**.** *Forecasts earn above-average weight in the global bagged combination if and only if they earn above-average weight within the subsets they appear in:*

$$\bar{v}_{k,i} > \frac{1}{m} \iff \mathbb{E}_S[w_{S,i}|i \in S] > \frac{1}{k}$$

This relationship between the weights has a natural interpretation. A forecast receives more than the equal global weight if and only if its mean weight, across the subsets in which it appears, exceeds $1/k$. Therefore, the global weight is dictated by its subset counterpart. This equivalence holds because the two benchmarks $\frac{1}{m}$ and $\frac{1}{k}$ differ by the inclusion probability under uniform rule $\frac{k}{m}$. The same quantity links global and conditional weights. Hence global importance is achieved locally. Alternatively, a forecast that is useful across different subsets is assigned a higher value at the global level.

Notice that both Proposition 2 and Proposition 3 require no assumptions on $\Sigma_{e,m}$ apart from it being a positive-definite matrix.

### 4.3 Common-Loading Heteroskedastic Forecast Error Model

We establish RSM in the general form without any additional assumptions on $\Sigma_{e,m}$. Every method (optimal weighting, averaging, RSM) relies on it since this matrix defines the loss through $Q^{opt}, Q^{ave}, Q_k^{bag,G}$, respectively. For infinite bagged RSM and average subset RSM, $\Sigma_{e.m}$ also enters in $Q_k^{bag,\infty}$ and $Q_k^{sub}$.

For further results we assume that forecast errors $u_t$ are characterized by a common-loading heteroskedastic structure. Formally:

**Assumption 3** (Forecast errors)**.** *Let forecast error be defined as follows:*

$$u_t = \lambda F_t \iota_m + \xi_t,$$

*where all forecasts have the same factor loading $\lambda$, $Var(\lambda F_t) = q > 0$ and $Var(\xi_t) = D = diag(\sigma_1^2, \ldots, \sigma_m^2)$, with at least one pair such that $\sigma_i^2 \neq \sigma_j^2$ for $i \neq j$, $0 < \tau_L \leq \tau_i \leq \tau_U < \infty$. $\mathbb{E}(\xi_t) = 0$, $Cov(F_t, \xi_t) = 0$.*

In other words, the forecast error vector $u_t$ can be decomposed into two quantities: a common factor that drives a shock to all $m$ forecasts with the same loading $\lambda$ and forecast-specific variation that is captured by $\xi_t$.

Assumption 3 in different variations is used in the literature. For instance, Lahiri and Sheng (2010) write forecast errors as the sum of a component common to all forecasters and idiosyncratic errors. Similarly, Elliott and Liao (2026), based on the SPF data, impose similar restrictions on the forecast error structure. A more general form with heterogeneous loadings and residual cross-sectional dependence can be found in Lee and Seregina (2020).

In this setup, the full forecast-error covariance matrix can be expressed as follows:

$$\Sigma_e = q\iota_m\iota_m' + D \tag{18}$$

The common loading assumption allows us to define the analogous matrix for a subset $S$ of size $k$. Since the common component $q$ is the same across all subsets, the forecast-error covariance matrix $\Sigma_{e,S}$ for a subset $S$ of size $k$ is described as follows:

$$\Sigma_{e,S} = q\iota_k\iota_k' + D_S, \quad D_S = diag(\sigma_i^2|i \in S), \tag{19}$$

where $D_S$ is the $k \times k$ principal submatrix of $D$ obtained by keeping only the columns and rows of forecasts that get drawn in a subset $S$. In the common loading case, $D_S$ is diagonal regardless of the subset taken.

To express further results more conveniently, let us introduce some additional notations.

**Definition 4** (Forecast precision)**.** *A forecast precision of a forecast $i$ is the inverse of its corresponding $\sigma^2$:*

$$\tau_i = \sigma_i^{-2} \implies D = diag\left(\frac{1}{\tau_i}, \cdots, \frac{1}{\tau_m}\right)$$

**Definition 5** (Sums of precisions)**.** *Suppose a subset $S$ is chosen uniformly at random as before. The sum of precisions over the subset $S$ can be expressed as follows:*

$$A_S = \sum_{i \in S} \tau_i$$

*Similarly, the sum of precisions over the full panel of $m$ forecasts has a following form*

$$A_m = \sum_{i=1}^{m} \tau_i \tag{20}$$

With the help of notations from Definition 4 and Definition 5, we can define the subset risk under common loading hetereskedastic forecast error structure as follows:

**Proposition 4** (Subset risk under common loading)**.** *For any subset $S$ of size $k$, the subset loss is as follows:*

$$Q(S) = q + \frac{1}{A_S} \tag{21}$$

*whereas the subset optimal weights are:*

$$w_{i,S} = \frac{\tau_i}{A_S}, \qquad i \in S. \tag{22}$$

The proof can be found in Appendix B.

The subset loss $Q(S)$ consists of two items: the irreducible common loading component $q$ and the inverse sum of forecast precisions drawn in the subset. The function $Q(S)$ is strictly decreasing with respect to the size of the subset $k$ for the nested subsets since $\tau_i > 0$ for all $i$. This means that by adding one may achieve a lower loss with the minimum value obtained by using $k = m$, that collapses to optimal weighting. This results once again shows the property of optimal weighting – its theoretical loss is the lowest possible. However, as discussed at the beginning of Section 4, the estimation error captured by the inflation factor $\mathcal{I}(T, m)$ potentially prevents it to produce the lowest loss in practice. Also, notice that optimal subset weight, unlike the subset loss, does not depend on $q$: it is a function of the forecast precision vector $\tau$.

### 4.4 Optimal and Equal Weighting under Common Loading

It is possible to define corresponding losses for the optimal weighting and equal weighting under the common loading heteroskedastic forecast errors in a similar way. The optimal weight for a forecast $i$ can be rewritten in terms of $\tau_i$ as follows:

$$w_i^{opt} = \frac{\sigma_i^{-2}}{\sum_{j=1}^{m} \sigma_j^{-2}} \implies w_i^{opt} = \frac{\tau_i}{A_m}$$

and its loss, $Q^{opt}$, as discussed in Section 4.3, has a similar form of the individual subset loss $Q(S)$. The only difference is that optimal weights uses all $m$ forecasts rather than $k$, which leads to the denominator of the second item be $A_m$ instead of $A_S$. That is why, $Q^{opt}$ has a following form:

$$Q^{opt} = q + \frac{1}{A_m}$$

On the other hand, the equal weighting produces the same weights as previously, that is $w^{ave} = \frac{1}{m}$. It does not take into account the heterogeneity across forecasts. By plugging the full forecast-error covariance matrix, as defined per Equation 18, we obtain the corresponding loss for equal weighting:

$$Q^{ave} = \frac{1}{m^2} \iota_m' \Sigma_e \iota_m = q + \frac{\sum_i \tau_i^{-1}}{m^2}$$

To illustrate the hidden similarity between the two losses, $Q^{opt}$ and $Q^{ave}$, let us denote $B = \frac{1}{\bar{\tau}}$ and $H = \overline{\sigma^2}\bar{\tau}$, where $\bar{\tau}$ is the mean precision and $\overline{\sigma^2}$ is the mean idiosyncratic variance.

Then, $B$ is the harmonic-mean idiosyncratic variance, or, alternatively, the reciprocal of average precision. It illustrates the scale of idiosyncratic noise and sets the overall level of the latter. Intuitively, it depicts how noisy a typical forecast is without taking into account heterogeneity. if forecasts are more precise, it leads to larger $\bar{\tau}$ value. This, in turn, decreases the value of $B$ and allows to achieve lower loss.

On the other hand, $H$ is a product of mean variance $\overline{\sigma^2}$ and mean precision $(\bar{\tau})$.It measures how spread out the idiosyncratic variances are. Notice, that $H$ is the ratio of the arithmetic mean to the harmonic mean of the variances. Thus, $H \geq 1$ with equality if and only if all $\sigma_i^2$ are equal to each other, which corresponds to the equicorrelation case, when equal weights are the optimal.

Let us rewrite both loss functions by plugging $B$ and $H$. In this case, we have:

$$Q^{opt} = q + \frac{B}{m}, \quad Q^{ave} = q + \frac{BH}{m}, \quad Q^{ave} - Q^{opt} = \frac{B(H-1)}{m}$$

This correspondence shows that the equal-weight loss can never beat the optimal weights in population, and the two coincide in the homogeneous case (all $\sigma_i^2$ are the same), thus forcing $H = 1$. Furthermore, it strengthens the reasoning of when RSM using $k > 1$ forecasts can outperform equal weighting – that is only when variances are dispersed enough such that the difference $Q^{ave} - Q^{opt}$ is substantial.

### 4.5 Average Subset RSM under Common Loading

Proposition 4 provides a subset loss under common loading. To derive a expected population loss of average subset RSM, the expectation over all $S$ needs to be taken.

**Proposition 5** (average subset risk under common loading)**.** *Using Taylor expansion, population average subset risk $Q_k^{sub}$ can be written as*

$$Q_k^{\mathrm{sub}} = \mathbb{E}_S[Q(S)] = q + \frac{1}{k\bar{\tau}} + \frac{m-k}{k^2(m-1)}\,\frac{v_\tau}{\bar{\tau}^3} + O\big(k^{-3}\big)\,, \tag{23}$$

*and, in leading order in $k$,*

$$Q_k^{\mathrm{sub}} = q + \frac{B}{k} + O\big(k^{-2}\big)\,, \tag{24}$$

*if the precisions are bounded away as mentioned in Assumption 3,*

The proof can be found in Appendix C.

Notice that the $Q_k^{sub}$ approximation stated in Proposition 5 looks similar to $Q^{opt}$ with the only difference being that the second item is divided by the subset size $k$ rather than by $m$. Therefore, inequality $Q_k^{sub} > Q^{opt}$ still holds in population. However, the estimation error that is captured by the inflation factor is less than the one used in the optimal case, i.e $\mathcal{I}(T,k) < \mathcal{I}(T,m)$.

We can write an optimization problem to find an optimal $k_{sub}^*$ which minimizes average subset expected population loss:

$$L_k^{sub} \approx \left(\sigma_\epsilon^2 + q + \frac{B}{k}\right)\left(\frac{T-1}{T-k}\right) \to min \tag{25}$$

which yields:

$$k^*_{sub} = \frac{-B + \sqrt{B^2 + (\sigma^2_\epsilon + q)BT}}{\sigma^2_\epsilon + q} > 0, \quad k^* = min\{min\{m, T-1\}, max\{1, round(k^*_{sub})\}\}$$

The proof can be found in Appendix D.

There is always a non-negative solution $k^*_{sub}$, which minimized the optimization function. Moreover, the optimal subset dimension depends scales roughly as $\sqrt{T}$ with $T$ being large compared to $B, \sigma^2_\epsilon$, and $q$. However, $k^*_{sub}$ needs to be clipped such that it is always between 1 and $m$. In other words, when $T$ is large enough such that $k^*_{sub} > m$, the full panel is to be used. The same logic applies to the case when $round(k^*_{sub}) < 1$.

### 4.6 Infinite Bagged RSM under Common Loading

Average subset risk evaluates the loss of a randomly selected subset before averaging across subset draws. On the contrary, infinite-bagged RSM averages the subset forecasts first and then evaluates the loss of the resulting combination. In this subsection, we analyze the infinite bagged RSM loss in the common loading case.

For a fixed subset $S$, the embedded subset weight is

$$v_{S,i} = \mathbf{1}\{i \in S\}\frac{\tau_i}{A_S}. \tag{26}$$

Therefore the infinite-bagged weight on forecast $i$ is

$$\bar{v}_{i,k} = \mathbb{E}_S[v_{S,i}] = \frac{k}{m}\tau_i\, \mathbb{E}_S\left[\frac{1}{\tau_i + \sum_{j\in R}\tau_j}\,\middle|\, i \in S\right], \tag{27}$$

The average-subset derivation in Appendix C, uses $\mathbb{E}_S[1/A_S]$, an unconditional expectation that has a clean representation. For the infinite bagged case, though, we need $\mathbb{E}_S[1/A_S \mid i \in S]$, the conditional expectation of $1/A_S$ given that forecasts $i$ is present.

When $i \in S$, the sum $A_S = \tau_i + A_{S\setminus\{i\}}$, where $A_{S\setminus\{i\}} = \sum_{j\in S, j\neq i}\tau_j$ is the sum of $k-1$ other precisions drawn from $\{\tau_j\}_{j\neq i}$. Conditioning on inclusion gives: $1/A_S$ around $A_{S\setminus\{i\}}$ yields:

$$\mathbb{E}_S\left[\frac{1}{A_S}\,\middle|\, i \in S\right] = \mathbb{E}\left[\frac{1}{\tau_i + A_{S\setminus\{i\}}}\right], \tag{28}$$

which is a harmonic mean shifted by $\tau_i$. This depends on the entire distribution of precision sums over the subsets of size $(k-1)$ of $\{\tau_j\}_{j\neq i}$ . There is no general closed form for this conditional harmonic mean without somehow restricting the profile $\{\tau_i\}$. The conditioning on $i \in S$ creates per-forecaster dependence that destroys this simplicity in comparison to the average subset case. However, it is possible to express $Q_k^{\text{bag},\infty}$ as follows. Since $\sum_i \bar{v}_{i,k} = 1$, the common factor contributes $q$ to the bagged risk. Thus:

$$Q_k^{\text{bag},\infty} = q + \sum_{i=1}^{m}\frac{\bar{v}^2_{i,k}}{\tau_i}. \tag{29}$$

This allows us to make a following proposition.

**Proposition 6** (RSM bagged risk representation)**.** *In the common-loading model,*

$$Q_k^{\text{bag},\infty} = Q^{\text{opt}} + R_k^{\text{bag}}, \tag{30}$$

*where*

$$R_k^{\text{bag}} = \sum_{i=1}^{m} \frac{(\bar{v}_{i,k} - w_i^{\text{opt}})^2}{\tau_i} \geq 0. \tag{31}$$

*or alternatively*

$$R_k^{bag} = \frac{1}{m^2} \sum_i \tau_i (g_i(k) - 1/\bar{\tau})^2 \tag{32}$$

*where*

$$g_i(k) = k\,\mathbb{E}_S \left[ \frac{1}{A_S} \,\middle|\, i \in S \right]. \tag{33}$$

The proof can be found in Appendix E.

By Equation 31, $R_k^{bag}$ is the squared distance, between the bagged weights $\bar{v}_{i,k}$ and the oracle weights $w^{opt}$. Moreover, the endpoints connect infinite bagged RSM to the standard combination rules. When $k = 1$, uniform subset selection produces $\bar{v}_{i,1} = \frac{1}{m}$ and the $R_1^{bag}$ illustrates the squared difference between the equal weighting and the optimal weighting schemes. This difference is small, when two weighting methods yield similar weights. In contrast, if $k = m$, the full subset selection implies $\bar{v}_{i,m} = w_i^{opt}$ and $R_m^{bag} = 0$. For intermediate cases $1 < k < m$, the infinite bagged RSM moves in between and the performance depends on the speed of convergence to the optimal weights.

The score representation, as presented in Equation 32 gives a useful interpretation of the infinite-bagged RSM weights. Since

$$\bar{v}_{i,k} = \frac{\tau_i}{m} g_i(k),$$

the quantity $g_i(k)$ can be interpreted as the conditional inverse-precision adjustment induced by drawing subsets that contain forecaster $i$. These scores satisfy the following property:

$$\sum_{i=1}^{m} \tau_i g_i(k) = m.$$

or equivalently,

$$\sum_{i=1}^{m} \tau_i \left( g_i(k) - \frac{1}{\bar{\tau}} \right) = 0.$$

Thus RSM does not change the total mass of the combination weights: the sum of the weights is always 1 regardless of the subset size $k$. However, it redistributes this mass across forecasts through the conditional harmonic terms $g_i(k)$.

In this score scale, the full oracle corresponds to the constant score:

$$g_i^{opt} = \frac{1}{\bar{\tau}},$$

because

$$w_i^{opt} = \frac{\tau_i}{m\bar{\tau}} = \frac{\tau_i}{m} g_i^{opt}.$$

Consequently, the excess-risk representation measures exactly how far the RSM scores $g_i(k)$ remain from the oracle score $1/\bar{\tau}$. The endpoints are also transparent. When $k = 1$, each selected subset contains only one forecaster, so $A_S = \tau_i$ conditional on $i \in S$, and therefore

$$g_i(1) = \frac{1}{\tau_i}.$$

This gives

$$\bar{v}_{i,1} = \frac{\tau_i}{m}\frac{1}{\tau_i} = \frac{1}{m},$$

so infinite-bagged RSM once again coincides with equal weighting. When $k = m$, the only possible subset is the full set, so $A_S = A_m = m\bar{\tau}$, and therefore

$$g_i(m) = \frac{k}{A_m} = \frac{1}{\bar{\tau}}.$$

Hence

$$\bar{v}_{i,m} = \frac{\tau_i}{m\bar{\tau}} = w_i^{opt}.$$

## 4.7 Bounds on $R_k$

The representation above shows that $R_k^{bag}$ is the squared distance between the infinite-bagged RSM weights and the oracle weights. Although the conditional harmonic means inside $g_i(k)$ do not admit a general closed form, the excess risk can still be bounded. Let

$$z_k = \frac{1}{k} - \frac{1}{m},$$

The exact bracket is useful because both endpoints can be expressed in the same coordinate $z_k = 1/k - 1/m$, which reveals the two boundary rates for the optimal subset size.

**Proposition 7** (Exact bracket for infinite-bagged excess risk)**.** *For every* $1 \le k \le m$*,*

$$\frac{(kE_S[1/A_S] - 1/\bar{\tau})^2}{\sum_{i=1}^m \sigma_i^2 - m/\bar{\tau}} \le R_k^{bag} \le E_S[1/A_S] - \frac{1}{A_m}.$$

*or, using Taylor expansion up to the leading order, as presented in Proposition 5,*

$$\frac{V_\delta^2}{m(H-1)\bar{\tau}} z_k^2 \lesssim R_k^{bag} \lesssim \frac{1}{\bar{\tau}} z_k, \tag{34}$$

*where* $V_\delta = \frac{v_\tau}{\bar{\tau}^2} = \frac{1}{m\bar{\tau}^2}\sum_{i=1}^m (\tau_i - \bar{\tau})^2$.

The proof can be found in Appendix F.

**Assumption 4** (Marginal regularity)**.** *Let*

$$\Delta R_k = R_k^{\mathrm{bag}} - R_{k+1}^{\mathrm{bag}}.$$

*For* $1 \ll k \ll m$*, suppose there exist constants* $0 < d_L \le d_U < \infty$*, independent of* $T$ *and* $k$*, such that*

$$d_L\left(z_k^2 - z_{k+1}^2\right) \le \Delta R_k \le d_U\left(z_k - z_{k+1}\right).$$

*Equivalently, for* $k \ll m$*,*

$$\Delta R_k \gtrsim k^{-3} \qquad \text{and} \qquad \Delta R_k \lesssim k^{-2}.$$

**Proposition 8** (Rate window for the optimal subset size). *Let*

$$L_k^{bag} \approx \left(\sigma_\epsilon^2 + Q^{opt} + R_k^{bag}\right) \frac{T-1}{T-k}.$$

*Let*

$$k_T^* \in \arg \min_{1 \le k \le min\{m, T-1\}} L_k^{bag}.$$

*Suppose Assumption 4 holds, $R_k^{bag}$ is uniformly bounded, and the pool size $m$ is large enough that the relevant range satisfies $T^{1/2} \ll m$. Then*

$$k_T^* = \Omega(T^{1/3}) \qquad \textit{and} \qquad k_T^* = O(T^{1/2}).$$

*Thus the optimal subset size lies between the cube-root and square-root rates:*

$$T^{1/3} \lesssim k_T^* \lesssim T^{1/2}.$$

*If $m$ is fixed or grows more slowly than $T^{1/2}$, the upper rate is truncated by the constraint $k \le m$.*

The proof can be found in Appendix G.

# 5 Monte Carlo Simulations

## 5.1 Monte Carlo Design

We evaluate the forecast-combination performance of RSM relative to the equal-weighted mean via Monte Carlo experiments. The simulation design follows Samuels and Sekkel (2017) and Alvarez et al. (2012). In all experiments, the number of predictors is fixed at $N = 50$. We run $P = 500$ replications and consider the time series length $T \in \{100, 500\}$.

The data generating process (DGP) for independent variables $x_t$ is specified as a dynamic factor structure:

$$x_t = F_t \Lambda + \epsilon_t, \tag{35}$$

where $\Lambda$ is a column vector of $N$ ones. $F_t$ is defined as an AR(1) process:

$$F_t = \phi_f F_{t-1} + \epsilon_t^f \qquad \epsilon_t^f \overset{iid}{\sim} N(0, 1)$$

We selected $\phi_f = \{0.3, 0.7\}$ to account for different persistence of $F_t$. Additionally, the error term in Equation 35 also follows the AR(1) process:

$$\epsilon_t = \phi_\eta \epsilon_{t-1} + \epsilon_t^\eta \qquad \epsilon_t^\eta \overset{iid}{\sim} N(0, \Sigma),$$

where the variance-covariance $\Sigma$ is given by Toeplitz matrix of the form $\Sigma_{ij} = \rho^{|i-j|}$ or alternatively:

$$\Sigma = \begin{pmatrix} 1 & \rho & \rho^2 & \cdots & \rho^{N-1} \\ \rho & 1 & \rho_s & \cdots & \rho^{N-2} \\ \rho^2 & \rho & 1 & \cdots & \rho^{N-3} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \rho^{N-1} & \rho^{N-2} & \rho^{N-3} & \cdots & 1 \end{pmatrix}.$$

The chosen form of $\Sigma$ allows us to introduce correlation into the idiosyncratic components across different predictors. We consider $\rho = \{0, 0.8\}$. Similar to $\phi_f$, we fix $\phi_\eta = \{0.3, 0.7\}$

Finally, the DGP for the dependent variable $y_{t+1}$ is defined as a linear combination of predictors $\{x_{it}\}_{i=1}^N$:

$$y_{t+1} = \beta' x_t + \varepsilon_{t+1} \qquad \varepsilon_{t+1} \overset{iid}{\sim} N(0,1)$$

The literature suggests multiple specifications for $\beta$ (Samuels and Sekkel, 2017). In this paper, we restrict our attention to a case where the sparsity is introduced. To be more specific, in the simulations $\beta = c[1_{1\times 10}, 0_{1\times(N-10)}]$ is assumed. Under this assumption, only the first ten variables – without loss of generality – have equal importance, whereas the rest of the predictors have zero loading in the DGP for $y_{t+1}$. Such sparsity is standard in macroeconomic applications, especially in forecasting.

Following Inoue and Kilian (2008), the scaling constant $c$ is chosen to match target population $R^2$ values of $\{0.1, 0.3, 0.5\}$. This parameterization spans low-, medium-, and high-signal environments for the dependent variable.

Based on each $x_{it}$, an out-of-sample prediction for $y_{t+1}$ is constructed with the help of a simple bivariate OLS regression:

$$y_{t+1} = \kappa_i x_{it} + \epsilon_{it}$$

Thus, the forecast pool consists of $N$ univariate (bivariate-in-estimation) forecasts, each using only one predictor $x_{it}$ (without an intercept), re-estimated with an expanding window at each origin. Later, these forecasts are combined via the RSM, the equal-weighted mean, or the optimal weighting and evaluated against the realized $y_{t+1}$. The RSM subset size is selected by cross-validation: we consider $k = \{2 \ldots N\}$, where if $n = k$, the estimation collapses to full OLS. Conditional on the selected $k$, we execute the RSM 1,000 times and average the resulting predictions.

The evaluation sample is $T_{eval} = 20$ periods, and $\kappa_i$ is estimated using an expanding window. We measure forecasting accuracy using Relative Root Mean Squared Forecast Error (rRMSFE), defined as follows:[5]

$$rRMSFE_{t+h} = \frac{\sqrt{\sum_{i=1}^N (y_{t+h} - \hat{y}_{t+h|t}^{RSM})^2}}{\sqrt{\sum_{i=1}^N (y_{t+h} - \hat{y}_{t+h|t}^{mean})^2}} = \frac{RMSFE_{t+h}^{RSM}}{RMSFE_{t+h}^{mean}}, \tag{36}$$

[5]Here, $h = 1$, as we focus on one-step-ahead forecasts in these simulations.

This metric facilitates comparison of the forecast-combination performance of the two approaches. A value below one indicates that RSM delivers more accurate forecasts than the standard equal-weighting scheme.

## 5.2 Monte Carlo Results

Figure 3 reports relative forecasting performance for $T = 100$, $\rho = 0.8$, $\phi_f = 0.3$, $\phi_\eta = 0.7$ first highlighting the role of $R^2$ in a small-sample setting. The vertical line corresponds to the $k = \sqrt{T}$ rounded to the nearest integer.

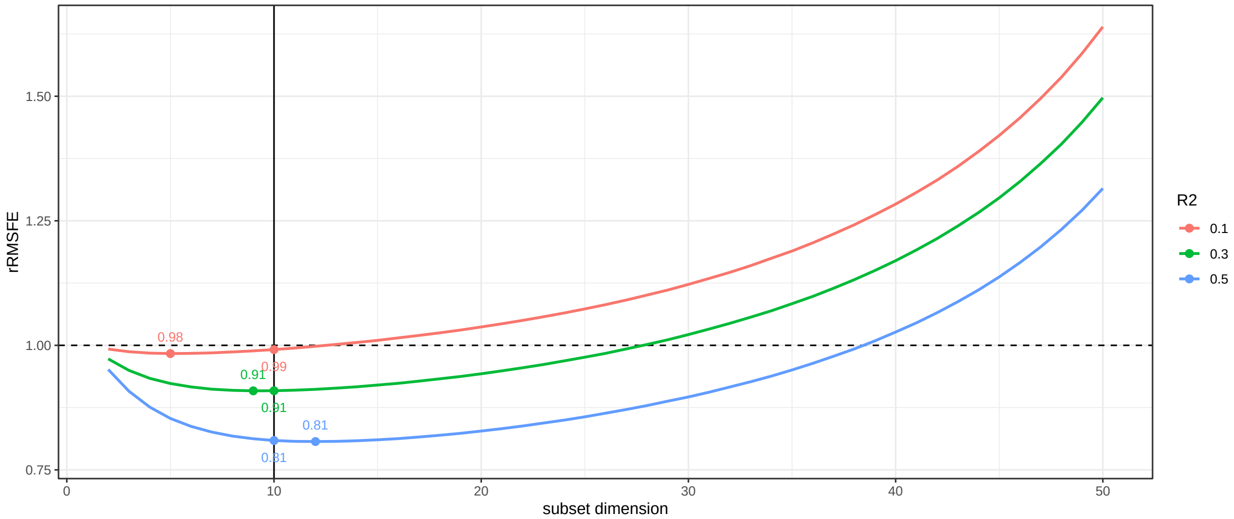


Figure 3: Relative forecasting performance ($T = 100$, $\rho = 0.8$, $\phi_f = 0.3$, $\phi_\eta = 0.7$)

The curves are (approximately) parallel, indicating that RSM's performance improves monotonically with $R^2$. This is unsurprising since higher values of $R^2$ imply that the predictors explain more of the target's variation. Accordingly, combining forecasts can improve accuracy.

It is worth noticing that for $R^2 \in \{0.3, 0.5\}$ the RSM dominates the equal-weighted mean. Their minimum values lie below one. By contrast, when $R^2 = 0.1$, gains are negligible: with a weak signal, individual forecasts have little predictive content, and their RSM aggregate is (almost) no better than the simple mean. Notice that by setting $k = \sqrt{T}$ the loss of gains is negligible compared to the true optimal subset size $k^*$. Those results are verified through all the scenarios, indicating that $\alpha \approx 1$ in the Equation **??**. The rest of the plots depicting the relative forecasting performance can be found in Appendix H.

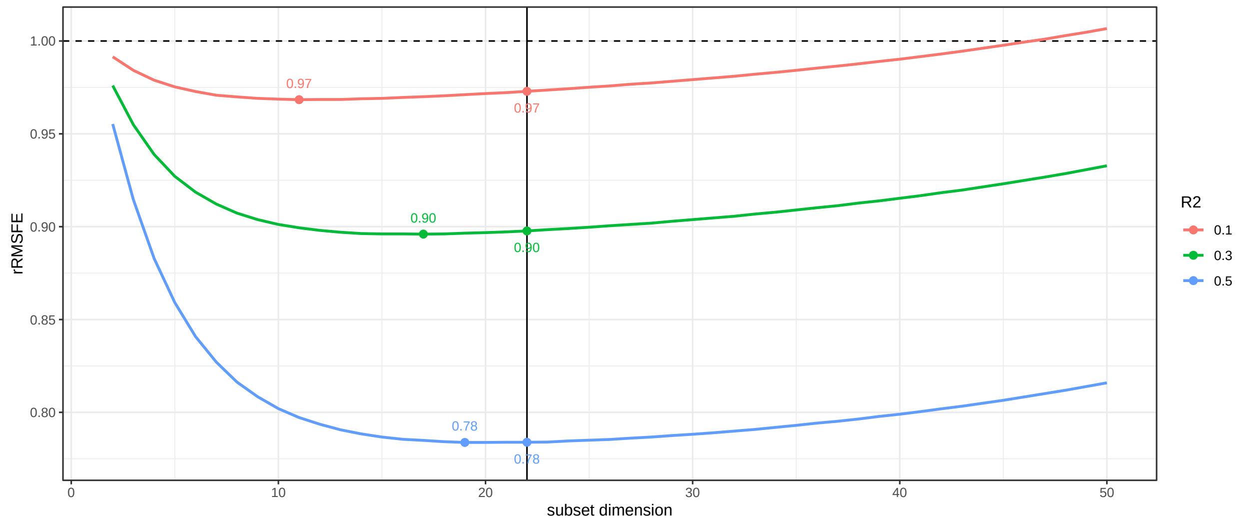


Figure 4: Relative forecasting performance ($T = 500$, $\rho = 0.8$, $\phi_f = 0.3$, $\phi_\eta = 0.7$)

Figure 4 presents relative forecasting performance for $T = 500$, $\rho = 0.8$, $\phi_f = 0.7$, $\phi_\eta = 0.3$. In other words, the effect of more data availability can be analyzed since all other parameters ($\rho$, $\phi_f$, $\phi_\eta$ remain unchanged. The larger sample size yields substantial gains from RSM over the equal-weighted mean, particularly at moderate and high $R^2$ values. When $R^2 = 0.5$, with other values fixed, the RSM can improve forecast combination results by almost 22%.

While tuning the optimal (or near-optimal) RSM subset size can be desirable for $T = 500$ and $R^2 \in \{0.3, 0.5\}$, a broad range of subset dimensions enhances forecasting accuracy (including $k = \sqrt{T}$). Across simulations, the optimal subset size either coincides with the rule-of-thumb choice or is nearby. Deviations from $k^*$ yield only negligible changes in forecast error relative to the exact optimum.

The Monte Carlo setup also allows us to evaluate $Q^{RSM}(k)$. Figure 5 depicts the latter for the case when $T = 500$, $\rho = 0.8$, $\phi_f = 0.3$, $\phi_\eta = 0.7$. For all other scenarios, $Q^{RSM}(k)$ is also depicted, and the corresponding plots could be found in Appendix H.1.

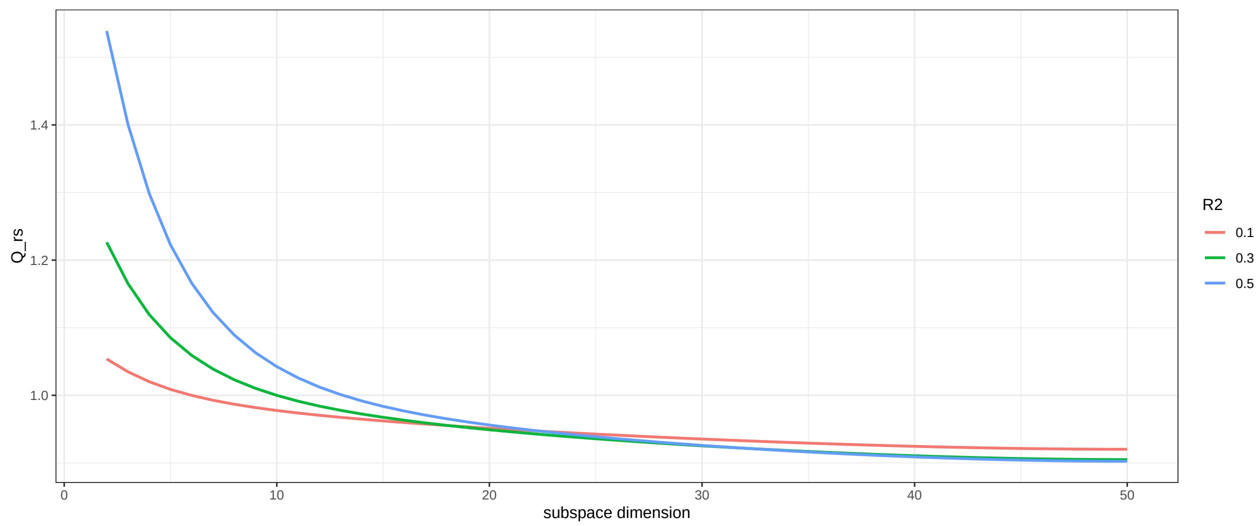


Figure 5: $Q^{RSM}$ ($T = 500$, $\rho = 0.8$, $\phi_f = 0.3$, $\phi_\eta = 0.7$)

As expected, the lowest value of $Q^{RSM}(k)$ is attained at $k = 50$, i.e., when all available predictions are included in the combination (and this holds in all other cases as well). However, the performance improvement relative to smaller subsets becomes very small once $k$ reaches moderate values. Overall, this result confirms that combining only a subset of $k$ forecasts typically incurs little additional loss compared with using all $N$ forecasts, while potentially reducing weight-estimation error.

# 6 Shapley Value Decomposition

One may explicitly pose a question of how important a given variable is to produce a predicted value of a target series. Interestingly, a possible answer to this question comes from a different field of study, namely game theory.

Shapley et al. (1953) introduced a solution to the problem of fairly dividing a joint pay-off across all individual players in a cooperative game. For almost six decades, this concept has had nothing to do with forecasting literature. This connection remained unexplored until Strumbelj and Kononenko (2010) established a connection between players and variables used for forecasting, as well as the pay-off and a final prediction.

Following Buckmann et al. (2022), it is possible to define a contribution of variable $k$ in observation $x_i$ and model $f$:

$$\phi_k^S(x_i; f) = \sum_{x' \subseteq \mathcal{C}(x) \setminus \{k\}} \frac{|x'|!(|N| - |x'| - 1)!}{|N|!} [f(x_i|x' \cup \{k\}) - f(x_i|x')],$$

where $\mathcal{C}(x) \setminus \{k\}$ corresponds to the set of all possible combinations of variables excluding $k$, $|x'|$ is the number of variables included in the combination.

Similar to the initial idea, the SVD represents a weighted sum of marginal contributions of a given variable (player) to all possible combinations excluding this variable. Note that we se the SVD for forecast combination. It means that a given variable, in this case, is an individual forecast. Thus, the contribution of an individual forecast to the final prediction is obtained.

Buckmann et al. (2022) described obstacles while estimating Equation 6. There are two major concerns. As stated earlier, the number of possible combinations with $l$ variables is $2^l$, thus it grows exponentially. With sufficiently large $l$, evaluation of all possible combinations seems infeasible. Additionally, estimating the term $f(x_i|x')$ is usually challenging (unless feature independence is assumed). Both of these problems, in principle, can be dodged if one opts for tree-based (see Lundberg et al. (2020)) or linear frameworks. In the following subsection, we describe the SVD in a linear setting and establish a bridge between the SVD and RSM.

## 6.1 SVD and RSM

In this subsection, we introduce the connection between RSM and the SVD. In a general forecast combination framework:

$$y_{T+h} = w_{T+h|T,0} + \boldsymbol{w'_{T+h|T}} \boldsymbol{\hat{y}}_{T+h|T} + \epsilon_{T+h}, \tag{37}$$

If optimal weights are used, then $w_{T+h|T,0} = 0$ and $\sum \boldsymbol{w'_{T+h|T}} = 1$. However, as shown further, both restrictions do not influence the SVD decomposition. In other words, practically, it does not matter whether to exploit the restricted or unrestricted OLS combined with RSM.

The contribution $\phi_j$ of a $j$-th forecast $\hat{y}_{T+h|T,j}$ on the total prediction $y_{T+h}$ is the difference between the feature effect and the average effect. More formally, one could express it as follows:

$$\phi_j = w'_{T+h|T,j} \hat{y}_{T+h|T,j} - E(w'_{T+h|T,j} X_j) = w'_{T+h|T,j} [\hat{y}_{T+h|T,j} - E(\hat{y}_{T+h|T,j})]$$

Let us combine this decomposition with RSM. It is straightforward to observe that in order to obtain $\phi_j$, one only misses the OLS coefficient (restricted or unrestricted) $w'_{T+h|T,j}$. However, in a high-dimensional case, directly obtaining $\beta_{x_j}$ may be complicated. RSM can assist in addressing this issue.

To illustrate the idea, suppose a variable $x_j$ was selected 50 times within RSM. This implies that one has 50 different estimations of $w'_{T+h|T,j}$, each based on different subsamples. However, a term $[\hat{y}_{T+h|T,j} - E(\hat{y}_{T+h|T,j})]$ remains the same across all the iterations. Thus, it is possible to obtain an averaged contribution of $\phi_j$ by averaging across different realizations of estimated $w'_{T+h|T,j}$, as depicted below:

$$\bar{\phi_j} = \hat{w'}_{T+h|T,j} [\hat{y}_{T+h|T,j} - E(\hat{y}_{T+h|T,j})], \tag{38}$$

where $\hat{w'}_{T+h|T,j}$ is the average of (restricted or unrestricted) OLS coefficients corresponding to different draws.

Recall that under a standard linear framework, the SVD possesses a valuable characteristic – the difference between the final prediction and the historical average prediction is fairly distributed among the covariates (individual forecasts). Maintaining this property is crucial when combining the SVD with RSM, as it is not inherently fulfilled by the combination by default.

The explanation lies in the following observation: $\hat{w'}_{T+h|T,j}$ and the final prediction are averaged across different numbers. That is, $\hat{\beta}_{x_j}$ is divided by the $n_{x_j}$ – the number of times a variable has been selected, meanwhile the final prediction is divided by $B$, where $B$ is the number of RSM iterations.

More formally, define inclusion probability and conditional mean weight:

$$p_j := \Pr(j \in S), \qquad m_j := \mathbb{E}[\,\beta_j \mid j \in S\,].$$

Additionally, define the weights: zero-bagged weight (including 0 when the forecast is not selected):

$$\bar{w}_j^{\text{zb}} := \mathbb{E}[\mathbf{1}\{j \in S\}\, w_j] \;=\; p_j\, m_j, \qquad \Rightarrow \quad \hat{y}_{t+1}^{RSM} = \sum_{i=1}^{N} \bar{w}_j^{\text{zb}} \hat{y}_{j,t+1}$$

and the conditional weight (only when the forecast is selected in a subsample $S$) which is equal to the conditional mean:

$$\bar{w}_j^{\text{nz}} := \mathbb{E}[w_j \mid j \in S] \;=\; m_j.$$

Thus, the average Shapley value for a $j$-th forecast uses the zero-bagged weight, that is:

$$\bar{\phi}_j = \bar{w}_j^{\text{zb}}[\hat{y}_{T+h|T,j} - E(\hat{y}_{T+h|T,j})] \tag{39}$$

However, the sum of the contributions defined in Equation 39 explains the difference between the model prediction and its average prediction. To explain this difference in terms of the dependent variable $(\hat{y}_{T+h} - \bar{y}_t)$, an additional intercept denoted $\phi_0$ should be added such that:

$$\hat{y}_{T+h} - \bar{y}_t = \phi_0 + \sum_j \phi_j, \tag{40}$$

where $\phi_0 = \sum_j \bar{w}_j^{zb} E[\hat{y}_{T+h|T,j}] - \bar{y}_t$ is the deviation between the model's average prediction and expected model prediction of the dependent variable, $\phi_j$ are the average Shapley values as defined in Equation 39. Notice that adding the constant term $\phi_0$ does not influence the estimation of $\phi_j$. Instead, it just ensures that the sum of $\phi_j$ explains the difference only in terms of the target variable $y_t$. Intuitively, the intercept term $\phi_0$ reconciles two baselines: the model's average fitted value (implied by averaged subset-specific combinations) and the unconditional mean of the realized target series.

This notation also helps to derive the key rule for comparing global $\bar{w}_j^{\text{zb}}$ and conditional $\bar{w}_j^{\text{nz}}$ weights to the equal weighting. That is, if $p_j = k/N$ is a uniform probability, then there is a correspondence between zero-bagged, no-zeros, and mean as follows:

$$\bar{w}_j^{\text{zb}} = \frac{k}{N} m_j \quad \Longrightarrow \quad \boxed{\bar{w}_j^{\text{zb}} > \tfrac{1}{N} \iff m_j > \tfrac{1}{k}} \tag{41}$$

This means that the zero-bagged weight is higher than the mean weight if the conditional weight is higher that the reciprocal of the subset dimension $k$. In other words, given that the variable is in the subset $S$, its expected weight should be higher than the mean weight in the subsample for the global weight to be higher than the mean.

# 7 Empirical Application

## 7.1 Data and Methods

We employ the FRED-MD dataset to nowcast GDP growth rates. Real-time vintages of the last months of each quarter (March, June, September, December) are used to produce a forecast. This vintage selection ensures the fullest data availability before actually observing the outcome. The number of time series in the dataset varies over time, usually rounding up to 120. The optimal subset space $k^*$ is set as proposed in the previous section, that is $k^*$ as the nearest integer to $\sqrt{T}$. The validation of the subset choice $k^*$ is presented in Appendix I.

We produce out-of-sample forecasts from Q3 1999 through Q1 2025, yielding 102 quarters for evaluation[6]. As in the Monte Carlo setup, we run separate bivariate OLS regressions —one for each predictor (aggregated to the quarterly frequency) — to obtain a unique forecast from each series. Thus, for a given quarter, we have as many forecasts as there are variables in the dataset. Current-quarter ragged-edge observations from the real time vintages (also aggregated to the quarterly frequency) are used to generate the out-of-sample nowcast from each predictor. At each vintage, the nowcast uses the latest available monthly releases within the quarter for each series; missing within-quarter observations are handled by the same aggregation rule used for all predictors to ensure comparability across vintages. To produce bivariate forecasts after the COVID-19-crisis (Q2 2020 and Q3 2020) as well as RSM and all the benchmarks forecasts, those two quarters were eliminated to avoid using outliers, which may severely affect the estimation. More specifically, for vintages after Q3 2020, we drop Q2 2020 and Q3 2020 from the estimation sample of every bivariate forecasting regression and from the RSM weight estimation step, while still evaluating forecasts on the full out-of-sample period unless explicitly stated otherwise. For the baseline scenario, we fix $k = \sqrt{T}$ rounded to the nearest integer.

Other benchmarks for high-dimensional forecast combination are estimated to compare the performance. Following Chen and Maung (2023), we use Random Forest, Lasso, and Ridge.

## 7.2 Forecast Combination Evaluation

Table 1 reports GDP nowcast combination results comparing RSM with the benchmarks for the full sample. RSM substantially outperforms the equal weighting, Ridge, and Random Forest in terms of the RMSFE. Lasso, on the other hand, produces similar results, thus being the closest competitor. However, the time span used for forecast evaluation includes the COVID-19 crisis, which can influence the nowcasting performance of all models.

| Model | MAFE | RMSFE |
|---|---|---|
| Mean | 0.54 | 1.10 |
| RSM | 0.45** | 0.74* |
| Lasso | 0.51 | 0.76* |
| Ridge | 0.50* | 1.00 |
| Random Forest | 0.55 | 1.24 |

Table 1: GDP Forecast Evaluation: Full Sample

[6]Due to missing values in some of the indicators, Q4 2003 is discarded.

Table 2 provides the same metrics without the two COVID-19 quarters (Q2 and Q3 2020). Excluding those quarters noticeably improves the forecast combination performance across the models. Still, RSM shows the best forecast combination performance (both in terms of MAFE and RMSFE). However, unlike the case with full data, both ridge and Random Forest appear to be more accurate than Lasso, the performance of which is the same as the equal weighting. This indicates that Lasso performed substantially better during the COVID-19 crisis than other competitors (except the RSM).

| Model | MAFE | RMSFE |
|---|---|---|
| Mean | 0.42 | 0.60 |
| RSM | $0.39^{**}$ | $0.53^{**}$ |
| Lasso | 0.45 | 0.60 |
| Ridge | $0.39^{**}$ | $0.55^{**}$ |
| Random Forest | $0.41^{**}$ | $0.58^{*}$ |

Table 2: GDP Forecast Evaluation: Excluding COVID-19

Recall, RSM takes an initial set of $N$ forecasts as input and uses them to produce another set of forecasts of size $B$, where $B$ is the number of iterations (in our case, $B = 1000$). Therefore, there is supposed to be a change in the distributions of forecasts, which helps RSM to predict the target better than the mean. However, this might happen only when both equal weighting and RSM produce substantially different results.

First, Figure 6 and Figure 7 depict both distributions for Q3 2004. This quarter is selected due to the similar prediction of the GDP growth rate for both methods[7]. Even though the RSM distribution is closer to the bell-shaped form, both are centered around the true realization without outliers. That is why taking the mean of both sets produces accurate estimations.

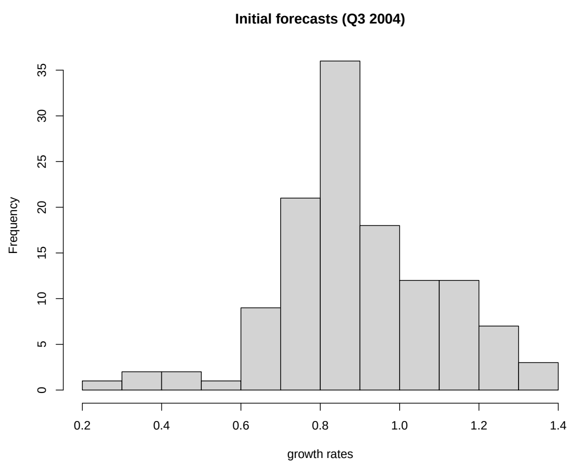


Figure 6: Initial set of GDP forecasts (Q3 2004)

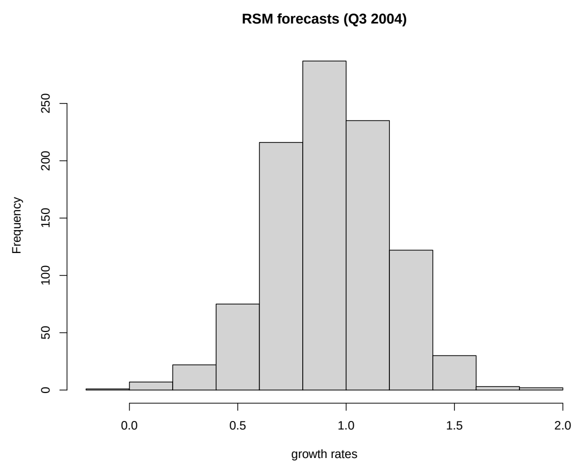


Figure 7: RS set of GDP forecasts (Q3 2004)

7

In crisis periods, RSM delivers significantly stronger out-of-sample performance (Figure 8 and Figure 9). Whereas many individual forecasts remain clustered near zero and fail to adjust, RSM adapts by shifting the combined forecast toward a more plausible mean level. Thus, RSM is particularly valuable when the target temporarily deviates from its typical behavior.

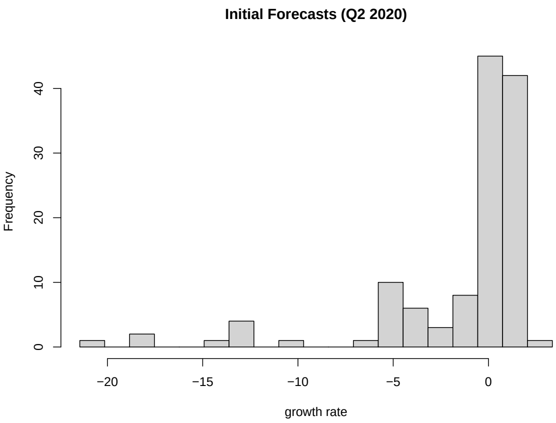


Figure 8: Initial set of GDP forecasts (Q2 2020)

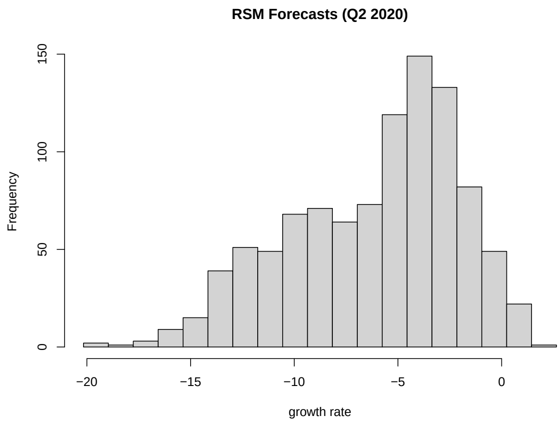


Figure 9: RS set of GDP forecasts (Q2 2020)

### 7.3 RSM Weights

RSM allows us to study the weight dynamics over time for any given variable and how it differs from the equal weighting scheme. Moreover, it is possible to study the global weight $\bar{w}_j^{zb}$ and its conditional counterpart $m_j$ as defined per Equation 41. In the main analysis, we stick to the global weight $\bar{w}_j^{zb}$ used to produce the final forecast. In Appendix J, a more detailed discussion of the correspondence between the global and conditional weights is presented.

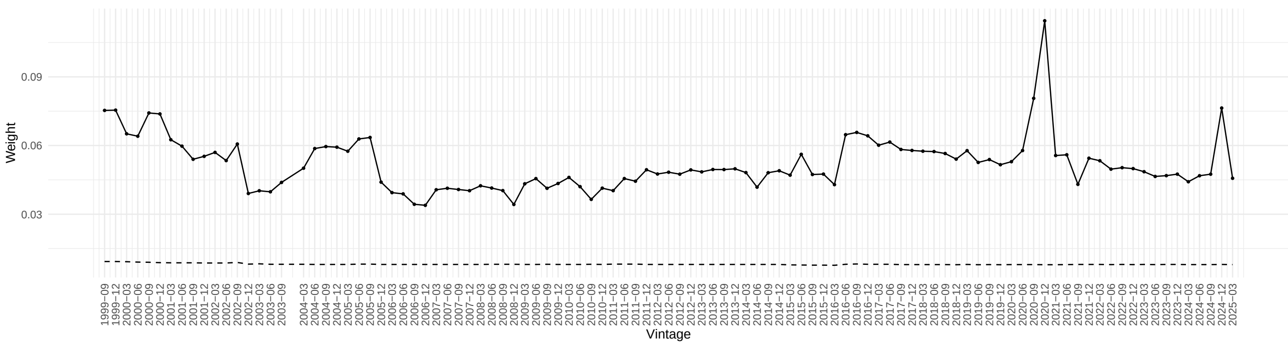


Figure 10: Industrial Production Weight over Time

We choose Industrial Production (FRED-md mnemonics INDPRO) to analyze the time-varying dynamics. This variable is chosen as one of the most important leading indicators used to now-cast/forecast GDP growth rates. IP is a standard real-activity indicator that tends to co-move with GDP and is widely used in short-horizon output tracking exercises. As shown in Figure 10, the solid line is always above the dotted one, meaning that Industrial Production always has a larger in absolute value than equal weight in the aggregated prediction. This confirms that forecasts based solely on IP are more informative and have stronger forecasting predictability than an average one, thus is supposed to have a higher weight in the optimal combination. Moreover, by implying the equal weighting, not only does one lose the flexibility, but also it influences the total relative contribution via the SVD as shown in Equation 38. Underestimating (as in this case) the weight of Industrial production leads to the corresponding underestimation of its contribution to the GDP growth rate which may negatively impact public policy decisions.

### 7.3.1 Negative Weights

In the forecast combination literature, a common assumption is that the weights should be non-negative and sum up to one in order to preserve the weight interpretability. The optimal weight, as defined in Equation 2, on the other hand, does not rely on the additional constraint of weights being non-negative. In other words, the optimal weights obtained via linear combination without additional assumptions can be negative. Moreover, if a large number of variables is included in the regression with the non-negativity constraints, it can lead to the case when most of the forecasts have a zero weight, which can underestimate their actual weight.

The negative weights can cause ambiguity in terms of interpretability. However, as shown in Radchenko et al. (2023), negative weights can arise (and it should be expected) in the case when forecasts to combine are highly correlated because of the multicollinearity. Since we do not impose a restriction of global weights to be non-negative, some of them in the final combination happened to be negative. For example, let us take the S&P 500 index (FRED-md mnemonics S_P_500), which is considered an important financial variable to predict the GDP growth rate Koop et al. (2021). In Figure 11 (upper plot), the global weight $\bar{w}^{zb}$ of the latter is depicted.

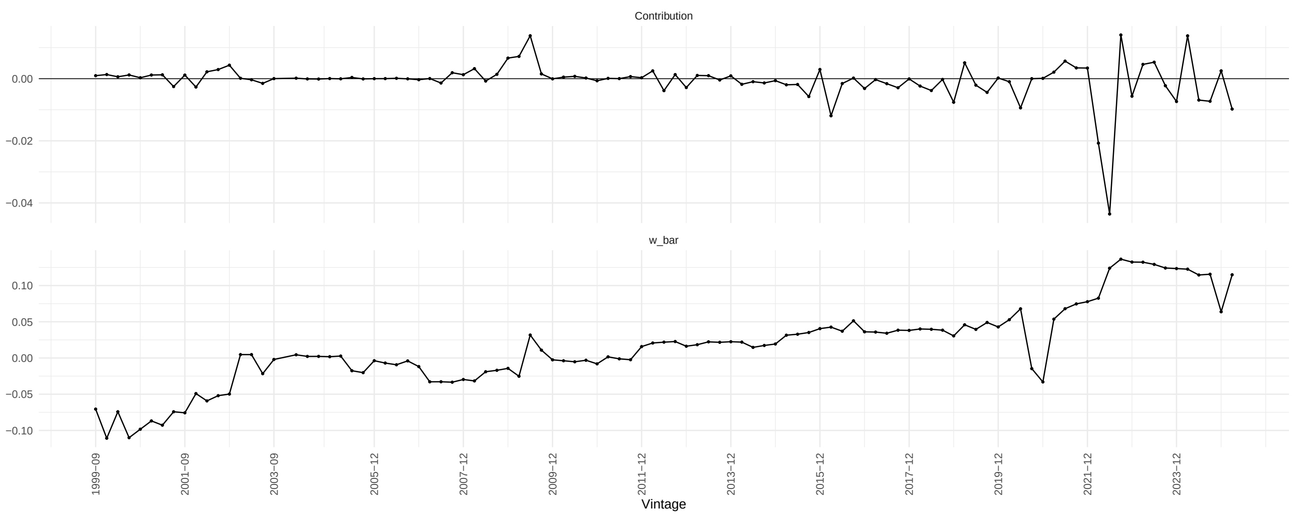


Figure 11: S&P 500: RSM global Weight and its Contribution

As can be seen from the plot, there are periods at the beginning of the out-of-sample exercise, when the RSM weights are indeed negative. It is worth mentioning that it happens quite often that some of the weights for different forecasts are less than 0 under the RSM forecast aggregation procedure. Again, according to Radchenko et al. (2023), it is expected to happen, since some of the employed forecasts in most subsets can be highly collinear, thus forcing the global weight to be negative.

We propose two alternative ways of interpreting negative weights within the RSM procedure. First, according to the Equation 37, when the simple OLS is estimated including a constant to capture a potential bias (if all the forecasts are unbiased, this constant is essentially 0), the obtained estimations are not interpreted as weights per se. Rather, they depict the OLS coefficients (the marginal change), and the focus is on the forecast combination performance instead of the interpretability of weights.

Second, we treat the weights as the input to compute the Shapley value of a given variable. This is, to the best of our knowledge, is an understudied property of the negative weights. The negative weights still allow for decomposing the contribution in such a way that the difference between the actual prediction and the historical average is fully explained. In other words, instead of focusing on the weight per se, we propose to focus on the Shapley value decomposition, where the (negative) weight serves as a definer. Because the combination assigns a negative weight to this forecast, its deviations from its own historical mean move the combined forecast in the opposite direction. A below-average forecast produces a positive contribution because the model is effectively subtracting that negative deviation.

In Figure 11 (lower plot), the corresponding attributions of the S&P 500 to the GDP growth rate are presented. Despite the negative weights at the start, its contribution was slightly positive, meaning that the deviation between the prediction and the historical average was negative. However, both the weight and the discrepancy being very close to 0 (even though both negative) result in the virtual absence of the contribution.

### 7.4 Shapley Value Decomposition

Once the linear decomposition values $\bar{\phi}_j$ are obtained for each variable, it is possible to plot them to observe the relative contributions. We choose the same quarters as in the Section 7.2, that is, Q3 2004 and Q2 2020.

In Figure 12 the ten highest (in absolute values) relative contributions are depicted for Q3 2004 as well as the sum of all other contributions, denoted "Other". This group aggregates the remaining forecasts' Shapley values (all contributions outside the top-10 in absolute value) so that the displayed bars still sum to the total explained deviation. As discussed in Section 7.2, in the given quarter, both the equal weighting and the RSM forecast combination methods provide similar results (0.89 and 0.94, respectively), essentially precisely predicting the outcome. Moreover, the actual growth rate of GDP in Q3 2004 (0.95) and the historical average of the latter (0.83) are very close to each other. Hence, the difference to be explained by the individual forecast is virtually small and $\phi_0 < 0.0007$. With the number of forecasts being high (124), a relative contribution of a given variable is expected to be close to 0.

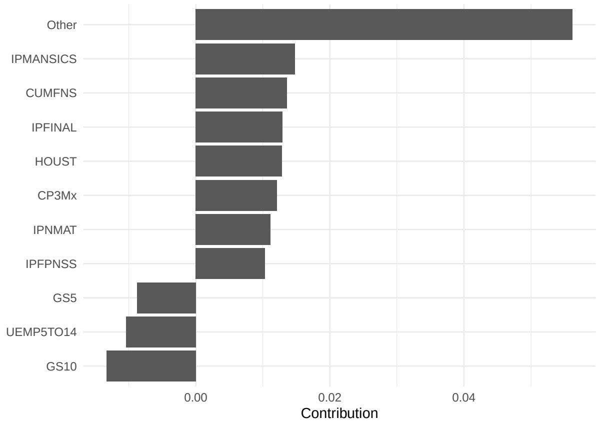


Figure 12: Shapley Value Decomposition (Q3 2004)

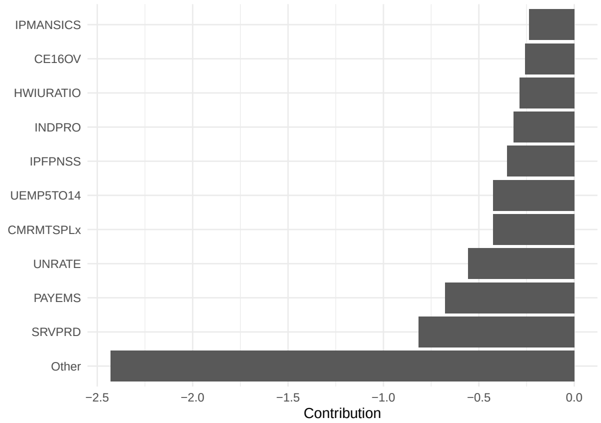


Figure 13: Shapley Value Decomposition (Q2 2020)

A more illustrative example is the same decomposition for Q2 2020, presented in Figure 13. Recall, RSM significantly outperforms the mean combiner for a given quarter in terms of forecast precision. More specifically, RSM predicts the GDP growth rate to be -6.1, whereas the mean combination yields -1.3, hence severely underestimates the true realization (-7.9). The estimated $\phi_0 = 0.032$.

Since the discrepancy between the RSM prediction and its historical average is substantial, the magnitude of each variable's Shapley value is higher than in normal times. This decomposition also allows for interpreting the results descriptively. The highest contributions have labor market variables alongside production.

### 7.4.1 RSM and Equal Weight Contributions

Not only does one oftentimes lose the forecasting precision by using the mean, but also the corresponding average Shapley values $\phi_j$ can be underestimated, which can negatively impact public policy decisions. As in Section 7.3, the Industrial production and S&P index are chosen to depict potential difference while estimating relative contributions using RSM weights and the equal weighting.

In Figure 14, average Shapley Values for Industrial production obtained by RSM and the equal weighting are depicted. Since, as discussed before, the individual contribution is usually around 0 almost by construction. Still, the difference between the lines is noticeable. Especially, it is clear in both the COVID-19 recession and the following rebound (Q2 2020 and Q3 2020, respectively). In both quarters, Industrial production has a much higher average Shapley Values compare to the mean.

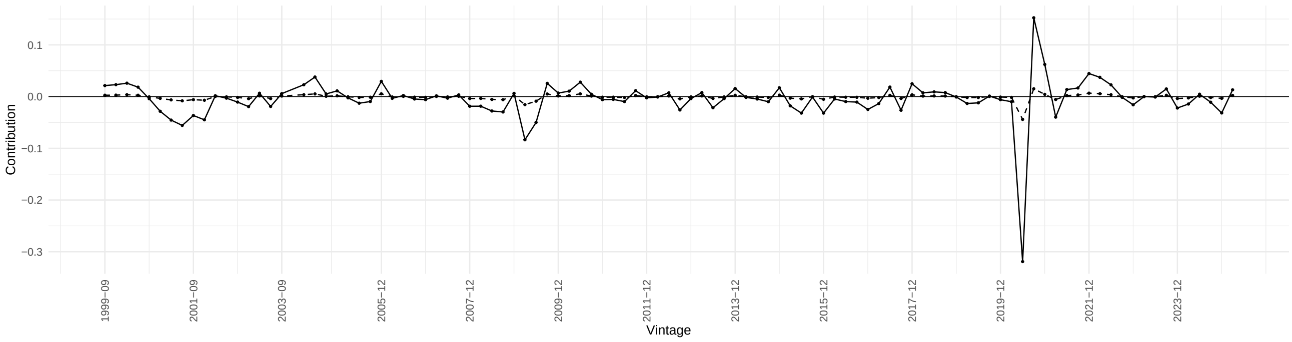


Figure 14: Industrial Production Contribution over Time

In Figure 15, average Shapley values obtained for S&P index by RSM and the equal weighting are depicted.

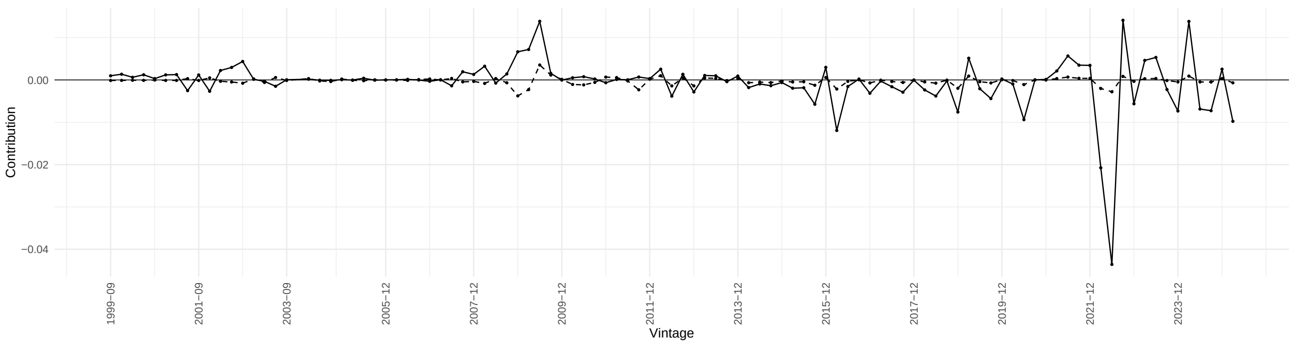


Figure 15: S& P Contribution over Time

# 8 Conclusion

This paper studies forecast aggregation in high-dimensional environments using RSM and its connection to the Shapley Value decomposition. The central idea is that instead of estimating a single large forecast-combination regression (often infeasible or noisy when the number of forecasts is large), we repeatedly draw small subsets of forecasts, estimate low-dimensional OLS combinations, and then average the resulting predictions. This strategy retains the information from large forecast pools while reducing estimation noise, which is a key friction behind the forecast combination puzzle and the weak empirical performance of many "optimal" weighting schemes.

Theoretical results show that there is room for RSM to improve forecast combination performance compared to the optimal weight and equal-weight frameworks in the finite (small) sample cases. Unsurprisingly, it is harder to beat the equal weighting because the latter does not imply the estimation error. Also, the performance of RSM crucially depends on the size of the subsets $k$. The ways of selecting $k$ in a general case and in some special cases are discussed. In line with the simulation evidence, a convenient and robust rule of thumb is to choose the subset size close to $k \approx \sqrt{T}$ rounded to the nearest integer, which tends to be near-optimal across designs and sample sizes.

The empirical application uses real-time vintages of the FRED-MD data (provided by the Federal Reserve Bank of St. Louis) to nowcast U.S. GDP growth, constructing a large set of univariate forecasts and then combining them. RSM consistently improves on the equal-weight mean and performs favorably relative to high-dimensional benchmarks such as Lasso, Ridge, and Random Forest. The gains are particularly pronounced in periods of unusual dynamics (notably the COVID-19 recession), when many individual predictors react unevenly and the simple mean can be pulled toward systematic under- or over-reaction.

Finally, the paper connects forecast combination with interpretability by showing how RSM can be paired with a Shapley value decomposition. RSM weights allow the model's prediction to be easily decomposed into contributions of individual forecasts. This provides an interpretable attribution even in high dimensions and is crucial for policymakers.

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# Appendices

## A Finite-bagging bridge

*Proof.* Let $V = v_S$ denote the random weight vector for a single subset draw, and $\bar{v} = \mathbb{E}[V]$ its expectation over all possible subsets. For independent draws, the finite-bagged estimator averages $G$ independent draws, meaning:

$$\hat{v} = \frac{1}{G}\sum_{b=1}^{G} V_b.$$

Then, by adding and subtracting $\bar{v}$ inside the quadratic form, we obtain that for a single realization:

$$\hat{v}'\Sigma_e\hat{v} = [\bar{v} + (\hat{v} - \bar{v})]'\Sigma_e[\bar{v} + (\hat{v} - \bar{v})]$$

After expanding the product on the right hand-side, we observe:

$$\hat{v}'\Sigma_e\hat{v} = \bar{v}'\Sigma_e\bar{v} + 2\bar{v}\Sigma_e(\hat{v} - \bar{v}) + (\hat{v} - \bar{v})'\Sigma_e(\hat{v} - \bar{v})$$

Taking expectation over the independent draws yields:

$$\mathbb{E}[\hat{v}'\Sigma_e\hat{v}] = \bar{v}'\Sigma_e\bar{v} + \mathbb{E}[(\hat{v} - \bar{v})'\Sigma_e(\hat{v} - \bar{v})].$$

Since the $\bar{v}'\Sigma_e\bar{v} = Q_k^{bag,\infty}$ is a constant, and the expectation of the cross term equals to zero since $\mathbb{E}(\hat{v} - \bar{v}) = 0$ because of $V_b$ being i.i.d draws from the same subset distribution. Therefore:

$$\mathbb{E}[\hat{v}'\Sigma_e\hat{v}] = Q_k^{bag,\infty} + \mathbb{E}[(\hat{v} - \bar{v})'\Sigma_e(\hat{v} - \bar{v})] \tag{42}$$

We can write the centered average explicitly as follows:

$$\hat{v} - \bar{v} = \frac{1}{G}\sum_{g=1}^{G}(V_g - \bar{v})$$

Substitute into the quadratic form and expand:

$$(\hat{v} - \bar{v})'\Sigma_e(\hat{v} - \bar{v}) = \frac{1}{G^2}\sum_{g=1}^{G}\sum_{g'=1}^{G}(V_g - \bar{v})'\Sigma_e(V_{g'} - \bar{v})$$

For the diagonal terms $g = g'$ due to $V_g$ being identically distributed to V, by taking expectation we get:

$$\mathbb{E}[(V_g - \bar{v})'\Sigma_{e,m}(V_g - \bar{v})] = \mathbb{E}[(V - \bar{v})'\Sigma_{e,m}(V - \bar{v})]$$

whereas for the non-diagonal terms $g \neq g'$ since $V_g$ and $V_{g'}$ are independent, for each pair of indices $(i, j)$ we obtain that their expectation is zero:

$$\mathbb{E}(V_g - \bar{v})'\Sigma_{e,m}(V_{g'} - \bar{v}) = \mathbb{E}(V_{g,i} - \bar{v}_i)\mathbb{E}(V_{g,j} - \bar{v}_j) = 0 \quad \forall g \neq g'$$

Thus:

$$\mathbb{E}[(\hat{v} - \bar{v})'\Sigma_{e,m}(\hat{v} - \bar{v})] = \frac{1}{G^2}G\mathbb{E}[(V - \bar{v})'\Sigma_{e,m}(V - \bar{v})] = \frac{1}{G}\mathbb{E}[(V - \bar{v})'\Sigma_{e,m}(V - \bar{v})]$$

Finally, expand and take expectation of the $(V - \bar{v})'\Sigma_{e,m}(V - \bar{v})$:

$$\mathbb{E}[(V - \bar{v})'\Sigma_{e,m}(V - \bar{v})] = \mathbb{E}(V'\Sigma_{e,m}V) - 2\bar{v}\Sigma_{e,m}E[V] + \bar{v}'\Sigma_e\bar{v} = Q_k^{sub} - Q_k^{bag,\infty}$$

and by plugging this expression back to Equation 42, we obtain:

$$Q_k^{\text{bag},G} = Q_k^{\text{bag},\infty} + \frac{1}{G}\left(Q_k^{\text{sub}} - Q_k^{\text{bag},\infty}\right).$$

□

# B Average subset under common loading heteroskedastic forecasts-errors

*Proof.* Recall $\Sigma_{e,S} = q\iota_k\iota_k' + D_S$. To invert it, we can use the Sherman-Morrison formula, which yields a following closed form:

$$\Sigma_{e,S}^{-1} = D_S^{-1} - \frac{qD_S^{-1}\iota_k\iota_k'D_S^{-1}}{1 + q\iota_k'D_S^{-1}\iota_k}. \tag{43}$$

Previously we define $A_S = \sum_{i\in S}\tau_i$, where $\tau_i = \sigma_i^{-2}$. Notice that $A_S$ can equivalently be written in terms of $D_S^{-1}$ as follows:

$$A_S = \iota_k'D_S^{-1}\iota_k = \sum_{i\in S}\tau_i.$$

Then by using Equation 43, we can write:

$$\iota_k'\Sigma_{e,S}^{-1}\iota_k = A_S - \frac{qA_S^2}{1+qA_S} = \frac{A_S}{1+qA_S}. \tag{44}$$

Finally, to find the subset risk take the reciprocal:

$$Q(S) = w_S'\Sigma_{e,S}w_S = (\iota_k'\Sigma_{e,S}^{-1}\iota_k)^{-1} = q + \frac{1}{A_S}. \tag{45}$$

To find optimal subset weights, first notice that:

$$\Sigma_{e,S}^{-1}\iota_k = \frac{D_S^{-1}\iota_k}{1+qA_S} = \frac{\boldsymbol{\tau}_S}{1+qA_S}, \tag{46}$$

where $\boldsymbol{\tau}_S$ is the $k\times 1$ vector of precisions of the forecasts in the subset $S$. By 46 into Equation 8, we obtain:

$$w_S = \frac{\Sigma_{e,S}^{-1}\iota_k}{\iota_k'\Sigma_{e,S}^{-1}\iota_k} = \frac{\boldsymbol{\tau}_S/(1+qA_S)}{A_S/(1+qA_S)} = \frac{\boldsymbol{\tau}_S}{A_S}.$$

Alternatively,

$$w_{i,S} = \frac{\tau_i}{A_S}$$

□

## C Average Subset RSM

*Proof.* By Proposition 4, each draw satisfies $Q(S) = q + \frac{1}{A_S}$. Taking the expectation over the uniform draw of $S$ yields:

$$Q_k^{\mathrm{sub}} = \mathbb{E}_S[Q(S)] = q + \mathbb{E}_S\left[\frac{1}{A_S}\right]. \tag{47}$$

Introduce the inclusion indicator $I_i = \mathbf{1}\{i \in S\}$, such that $A_S = \sum_{i=1}^m I_i \tau_i$. Under uniform sampling without replacement at random, the first- and second-order inclusion probabilities are as follows:

$$\mathbb{P}(i \in S) = \frac{k}{m}, \qquad \mathbb{P}(i, j \in S) = \frac{\binom{m-2}{k-2}}{\binom{m}{k}} = \frac{k(k-1)}{m(m-1)}, \quad i \neq j.$$

Hence:

$$\mathbb{E}[I_i] = \frac{k}{m}, \qquad \mathrm{Var}(I_i) = \frac{k}{m}\left(1 - \frac{k}{m}\right),$$

and, for $i \neq j$,

$$\mathrm{Cov}(I_i, I_j) = \frac{k(k-1)}{m(m-1)} - \frac{k^2}{m^2} = -\frac{k(m-k)}{m^2(m-1)}.$$

By linearity, the mean of $A_S$ is

$$\mathbb{E}[A_S] = \sum_{i=1}^m \tau_i\,\mathbb{E}[I_i] = \frac{k}{m}\sum_{i=1}^m \tau_i = k\bar{\tau}. \tag{48}$$

For the variance,

$$\mathrm{Var}(A_S) = \sum_{i=1}^m \tau_i^2\,\mathrm{Var}(I_i) + \sum_{i\neq j} \tau_i\tau_j\,\mathrm{Cov}(I_i, I_j) = \frac{k(m-k)}{m^2}\sum_{i=1}^m \tau_i^2 - \frac{k(m-k)}{m^2(m-1)}\sum_{i\neq j}\tau_i\tau_j.$$

Using $\sum_{i\neq j}\tau_i\tau_j = \left(\sum_i \tau_i\right)^2 - \sum_i \tau_i^2 = A_m^2 - \sum_i \tau_i^2$ with $A_m = m\bar{\tau}$,

$$\mathrm{Var}(A_S) = \frac{k(m-k)}{m^2(m-1)}\left[(m-1)\sum_i \tau_i^2 - A_m^2 + \sum_i \tau_i^2\right] = \frac{k(m-k)}{m^2(m-1)}\left[m\sum_i \tau_i^2 - A_m^2\right].$$

Since $m\sum_i \tau_i^2 - A_m^2 = m\sum_i \tau_i^2 - m^2\bar{\tau}^2 = m\sum_i(\tau_i - \bar{\tau})^2 = m^2 v_\tau$, this simplifies to the hypergeometric variance:

$$\mathrm{Var}(A_S) = \frac{k(m-k)}{m-1}\,v_\tau. \tag{49}$$

Write $\mu = \mathbb{E}[A_S] = k\bar{\tau}$ and $\epsilon_S = A_S - \mu$, so $\mathbb{E}[\epsilon_S] = 0$ and $\mathbb{E}[\epsilon_S^2] = \mathrm{Var}(A_S)$. Since $A_S \geq k\tau_{\min}$, Taylor's theorem to third order gives the following expansion:

$$\frac{1}{A_S} = \frac{1}{\mu} - \frac{\epsilon_S}{\mu^2} + \frac{\epsilon_S^2}{\mu^3} - \frac{\epsilon_S^3}{\mu^4} + \rho_S, \qquad |\rho_S| \leq \frac{|\epsilon_S|^4}{(k\tau_{\min})^5}.$$

Under the uniform boundedness condition for the precision profile and simple random sampling without replacement,

$$\mathbb{E}[\epsilon_S^3] = O(k), \qquad \mathbb{E}[\epsilon_S^4] = O(k^2).$$

Hence

$$-\frac{\mathbb{E}[\epsilon_S^3]}{\mu^4} = O(k^{-3}), \qquad \mathbb{E}[\rho_S] = O(k^{-3}),$$

so

$$\mathbb{E}_S\left[\frac{1}{A_S}\right] = \frac{1}{\mu} + \frac{Var(A_S)}{\mu^3} + O(k^{-3}). \tag{50}$$

Substituting $\mu = k\bar{\tau}$ and (49) into (50),

$$\mathbb{E}_S\left[\frac{1}{A_S}\right] = \frac{1}{k\bar{\tau}} + \frac{1}{(k\bar{\tau})^3} \cdot \frac{k(m-k)}{m-1}\, v_\tau + O\big(k^{-3}\big) = \frac{1}{k\bar{\tau}} + \frac{m-k}{k^2(m-1)}\, \frac{v_\tau}{\bar{\tau}^3} + O\big(k^{-3}\big)\,.$$

Combining with (47) yields the second-order expansion (23). Since the correction term is $O(k^{-2})$, writing $B = 1/\bar{\tau}$ gives the leading form (24) □

## D Average Subset RSM – Minimization Problem

Applying the finite-sample inflation factor $\mathcal{I}(T,k) = \frac{T-1}{T-k}$, which accounts for estimating $k$ combination weights from $T$ observations, the total average-subset loss is

$$L_k^{\text{sub}} \approx \left(\sigma_\epsilon^2 + Q_k^{\text{sub}}\right) \mathcal{I}(T,k).$$

Substituting the leading-order expansion $Q_k^{\text{sub}} = q + \frac{B}{k} + O(k^{-2})$ of Proposition 5, with $B = 1/\bar{\tau}$, gives

$$L_k^{\text{sub}} \approx \left(\sigma_\epsilon^2 + q + \frac{B}{k}\right) \frac{T-1}{T-k}. \tag{51}$$

The optimal subset size solves the (continuous-relaxation) program

$$k_{\text{sub}}^* = \arg \min_{1 \le k \le min\{m, T-1\}} \left(\sigma_\epsilon^2 + q + \frac{B}{k}\right) \frac{T-1}{T-k}. \tag{52}$$

**Proposition 9** (Square-root rule for the average-subset estimator)**.** *Under the common-loading structure, the minimizer of* (51) *over* $k^* = min\{min\{m, T-1\}, max\{1, round(k_{sub}^*)\}\}$ *is the positive root of*

$$(\sigma_\epsilon^2 + q)\, k^2 + 2B\, k - B\, T = 0, \tag{53}$$

*namely*

$$k_{\text{sub}}^* = \frac{-B + \sqrt{B^2 + (\sigma_\epsilon^2 + q)\, B\, T}}{\sigma_\epsilon^2 + q} \sim \sqrt{\frac{B\, T}{\sigma_\epsilon^2 + q}} = \sqrt{\frac{T}{\bar{\tau}\,(\sigma_\epsilon^2 + q)}} \propto \sqrt{T}. \tag{54}$$

*Proof.* Differentiate (51) with respect to $k$. With $f(k) = \sigma_\epsilon^2 + q + B/k$ and $h(k) = T - k$, so that $L_k^{\text{sub}} = (T-1)\, f/h$, the quotient rule gives

$$\frac{\mathrm{d}L_k^{\text{sub}}}{\mathrm{d}k} = (T-1)\, \frac{f'(k)\, h(k) - f(k)\, h'(k)}{h(k)^2} = (T-1)\, \frac{-\frac{B}{k^2}(T-k) + \left(\sigma_\epsilon^2 + q + \frac{B}{k}\right)}{(T-k)^2}.$$

Setting the numerator to zero and multiplying by $k^2$,

$$-B(T-k) + (\sigma_\epsilon^2 + q)k^2 + Bk = 0 \quad \Longleftrightarrow \quad (\sigma_\epsilon^2 + q)k^2 + 2Bk - BT = 0,$$

which is (53). The quadratic has one sign change in its coefficients ($B, T > 0$), hence a unique positive root,

$$k_{\text{sub}}^* = \frac{-2B + \sqrt{4B^2 + 4(\sigma_\epsilon^2 + q)BT}}{2(\sigma_\epsilon^2 + q)} = \frac{-B + \sqrt{B^2 + (\sigma_\epsilon^2 + q)BT}}{(\sigma_\epsilon^2 + q)}.$$

Because $L_k^{\text{sub}} \to \infty$ both as $k \downarrow 0$ (through the $B/k$ term) and as $k \uparrow T$ (through $\mathcal{I}(T,k)$), this interior stationary point is the global minimizer on $(0, T)$. Finally, for large $T$ the constant and linear terms are negligible relative to $(\sigma_\epsilon^2 + q)BT$, so $k_{\text{sub}}^* = \sqrt{BT/(\sigma_\epsilon^2 + q)}\,\left(1 + O(T^{-1/2})\right)$, giving the $\sqrt{T}$ scaling in (54). □

## E RSM bagged risk representation

*Proof.* Under the common-loading structure,

$$\Sigma_e = q\iota_m\iota_m' + D, \qquad D = \mathrm{diag}(\sigma_1^2, \dots, \sigma_m^2),$$

where $\tau_i = \sigma_i^{-2}$. Since both the oracle weights $w^{opt}$ and the infinite-bagged weights $\bar{v}_k$ sum to one, the common component contributes the same amount $q$ to both risks. Therefore,

$$Q_k^{bag,\infty} = q + \sum_{i=1}^{m} \frac{\bar{v}_{i,k}^2}{\tau_i}, \qquad Q^{opt} = q + \sum_{i=1}^{m} \frac{(w_i^{opt})^2}{\tau_i}.$$

Let

$$\delta_i = \bar{v}_{i,k} - w_i^{opt}.$$

Because both weight vectors sum to one,

$$\sum_{i=1}^{m} \delta_i = 0.$$

Moreover, under common loading the oracle weights are

$$w_i^{opt} = \frac{\tau_i}{A_m},$$

Hence

$$\frac{w_i^{opt}}{\tau_i} = \frac{1}{A_m}.$$

Now expand the idiosyncratic component of the infinite-bagged risk around the oracle weights:

$$\sum_{i=1}^{m} \frac{\bar{v}_{i,k}^2}{\tau_i} = \sum_{i=1}^{m} \frac{(w_i^{opt} + \delta_i)^2}{\tau_i}.$$

This gives

$$\sum_{i=1}^{m} \frac{\bar{v}_{i,k}^2}{\tau_i} = \sum_{i=1}^{m} \frac{(w_i^{opt})^2}{\tau_i} + 2\sum_{i=1}^{m} \frac{w_i^{opt}\delta_i}{\tau_i} + \sum_{i=1}^{m} \frac{\delta_i^2}{\tau_i}.$$

The cross term is zero because

$$2\sum_{i=1}^{m} \frac{w_i^{opt}\delta_i}{\tau_i} = \frac{2}{A_m}\sum_{i=1}^{m} \delta_i = 0.$$

Therefore,

$$Q_k^{bag,\infty} = Q^{opt} + \sum_{i=1}^{m} \frac{(\bar{v}_{i,k} - w_i^{opt})^2}{\tau_i}.$$

This proves

$$Q_k^{bag,\infty} = Q^{opt} + R_k^{bag}, \qquad R_k^{bag} = \sum_{i=1}^{m} \frac{(\bar{v}_{i,k} - w_i^{opt})^2}{\tau_i} \geq 0.$$

Let us derive the alternative representation. Recall, for a subset $S$, the embedded subset-optimal weight assigned to forecaster $i$ is

$$v_{S,i} = \mathbf{1}\{i \in S\}\frac{\tau_i}{A_S},$$

Hence the infinite-bagged weight is

$$\bar{v}_{i,k} = E_S[v_{S,i}] = P(i \in S)\tau_i E_S\left[\frac{1}{A_S} \mid i \in S\right].$$

Since subsets are drawn uniformly with fixed size $k$,

$$P(i \in S) = \frac{k}{m}.$$

Using the definition

$$g_i(k) = kE_S\left[\frac{1}{A_S} \mid i \in S\right],$$

we obtain

$$\bar{v}_{i,k} = \frac{\tau_i}{m} g_i(k).$$

On the other hand,

$$w_i^{opt} = \frac{\tau_i}{A_m} = \frac{\tau_i}{m\bar{\tau}}.$$

Therefore,

$$\bar{v}_{i,k} - w_i^{opt} = \frac{\tau_i}{m}\left(g_i(k) - \frac{1}{\bar{\tau}}\right).$$

Substituting this expression into the previous formula for $R_k^{bag}$ gives

$$R_k^{bag} = \sum_{i=1}^{m} \frac{1}{\tau_i}\left[\frac{\tau_i}{m}\left(g_i(k) - \frac{1}{\bar{\tau}}\right)\right]^2.$$

Thus,

$$R_k^{bag} = \frac{1}{m^2}\sum_{i=1}^{m} \tau_i \left(g_i(k) - \frac{1}{\bar{\tau}}\right)^2.$$

□

## F $R_k^{bag}$ bounds

*Proof.* The upper bound follows from Jensen's inequality as defined in Equation 15. Subtracting $Q^{opt}$ and using

$$Q_k^{sub} = q + E_S[1/A_S], \qquad Q^{opt} = q + \frac{1}{A_m},$$

gives

$$R_k^{bag} \leq E_S[1/A_S] - \frac{1}{A_m}.$$

For the lower bound, recall the score representation

$$R_k^{bag} = \frac{1}{m^2} \sum_{i=1}^{m} \tau_i \left( g_i(k) - \frac{1}{\bar{\tau}} \right)^2.$$

Define

$$h_i(k) = g_i(k) - \frac{1}{\bar{\tau}}.$$

The score identities imply

$$\sum_{i=1}^{m} \tau_i g_i(k) = m, \qquad \sum_{i=1}^{m} g_i(k) = kmE_S[1/A_S].$$

Therefore,

$$\sum_{i=1}^{m} \tau_i h_i(k) = \sum_{i=1}^{m} \tau_i g_i(k) - \frac{1}{\bar{\tau}} \sum_{i=1}^{m} \tau_i = m - \frac{A_m}{\bar{\tau}} = 0,$$

because $A_m = m\bar{\tau}$. Also,

$$\sum_{i=1}^{m} h_i(k) = kmE_S[1/A_S] - \frac{m}{\bar{\tau}} = m \left( kE_S[1/A_S] - \frac{1}{\bar{\tau}} \right).$$

Let

$$d_k = m \left( kE_S[1/A_S] - \frac{1}{\bar{\tau}} \right).$$

We now minimize

$$\sum_{i=1}^{m} \tau_i h_i^2$$

over all vectors $h = (h_1, \ldots, h_m)$ satisfying

$$\sum_i \tau_i h_i = 0, \qquad \sum_i h_i = d_k.$$

Since the actual vector $h_i(k)$ satisfies these restrictions, its weighted norm cannot be smaller than this constrained minimum.

To compute the minimum, use the Lagrangian

$$\mathcal{L}(h, \lambda, \mu) = \sum_{i=1}^{m} \tau_i h_i^2 + \lambda \sum_{i=1}^{m} \tau_i h_i + \mu \left( \sum_{i=1}^{m} h_i - d_k \right).$$

The first-order condition is

$$2\tau_i h_i + \lambda \tau_i + \mu = 0,$$

so

$$h_i = a + \frac{b}{\tau_i}$$

for constants $a, b$. The restriction $\sum_i \tau_i h_i = 0$ gives

$$aA_m + bm = 0, \qquad \text{so} \qquad a = -\frac{bm}{A_m} = -\frac{b}{\bar{\tau}}.$$

Hence

$$h_i = b\left(\frac{1}{\tau_i} - \frac{1}{\bar{\tau}}\right).$$

Using $\sum_i h_i = d_k$,

$$b\left(\sum_{i=1}^{m} \frac{1}{\tau_i} - \frac{m}{\bar{\tau}}\right) = d_k.$$

Thus

$$b = \frac{d_k}{\sum_i 1/\tau_i - m/\bar{\tau}}.$$

The minimum weighted norm is therefore

$$\sum_i \tau_i h_i^2 = b^2 \sum_i \tau_i \left(\frac{1}{\tau_i} - \frac{1}{\bar{\tau}}\right)^2.$$

The last sum simplifies to

$$\sum_i \tau_i \left(\frac{1}{\tau_i} - \frac{1}{\bar{\tau}}\right)^2 = \sum_i \frac{1}{\tau_i} - \frac{m}{\bar{\tau}}.$$

Therefore,

$$\sum_i \tau_i h_i^2 \geq \frac{d_k^2}{\sum_i 1/\tau_i - m/\bar{\tau}}.$$

Since $1/\tau_i = \sigma_i^2$ and

$$R_k^{bag} = \frac{1}{m^2} \sum_i \tau_i h_i(k)^2,$$

we obtain

$$R_k^{bag} \geq \frac{(kE_S[1/A_S] - 1/\bar{\tau})^2}{\sum_i \sigma_i^2 - m/\bar{\tau}}.$$

Now, let us show that those bounds can be simplified by using Taylor expansion up to the leading order, as per Proposition 5. For the upper bound,

$$\mathbb{E}_S[1/A_S] - \frac{1}{A_m} = \frac{1}{k\bar{\tau}} - \frac{1}{m\bar{\tau}} + o(z_k) = \frac{1}{\bar{\tau}} z_k + o(z_k).$$

Thus the Jensen ceiling is linear in $z_k$:

$$R_k^{\text{bag}} \lesssim \frac{1}{\bar{\tau}} z_k.$$

For the lower bound, define

$$D_k = k\,\mathbb{E}_S[1/A_S] - \frac{1}{\bar{\tau}}.$$

The first-order term cancels, and the Taylor expansion gives

$$D_k = \frac{\mathrm{Var}_S(A_S)}{k^2 \bar{\tau}^3} + o(z_k) = \frac{m}{m-1} \frac{V_\delta}{\bar{\tau}} z_k + o(z_k).$$

Also,

$$\sum_i \sigma_i^2 - \frac{m}{\bar{\tau}} = \frac{m(H-1)}{\bar{\tau}},$$

where

$$H = \overline{\sigma^2}\,\bar{\tau}.$$

Therefore the projection floor is quadratic in $z_k$:

$$R_k^{\text{bag}} \gtrsim \frac{V_\delta^2}{m(H-1)\bar{\tau}} z_k^2,$$

up to the finite-$m$ factor $m^2/(m-1)^2$.

Hence, to leading order,

$$\frac{V_\delta^2}{m(H-1)\bar{\tau}} z_k^2 \lesssim R_k^{\text{bag}} \lesssim \frac{1}{\bar{\tau}} z_k.$$

□

## G Rates

*Proof.* We now compute the two boundary optimizers implied by the leading order bounds stated in Appendix F.

The Jensen ceiling leads to the same objective studied in Appendix D, and therefore gives the square-root rule,

$$k_U^* \asymp T^{1/2}.$$

For the lower boundary, set

$$R_k^L = C_L z_k^2, \qquad C_L = \frac{V_\delta^2}{m(H-1)\bar{\tau}}.$$

The continuous first-order condition is

$$-\frac{2C_L z_k}{k^2}(T-k) + A + C_L z_k^2 = 0.$$

For the interior range $k \ll m$, so that $z_k \approx 1/k$, this becomes

$$Ak^3 + 3C_L k - 2C_L T = 0.$$

Hence

$$k_L^* \sim \left(\frac{2C_L T}{A}\right)^{1/3} = \left(\frac{2V_\delta^2 T}{m(H-1)\bar{\tau}A}\right)^{1/3}.$$

Thus the projection floor gives the cube-root rule,

$$k_L^* \asymp T^{1/3}.$$

The level bounds alone do not mechanically imply that the minimiser of the true $R_k^{\mathrm{bag}}$ lies between these two optima. For that conclusion we require a marginal regularity condition.

Let

$$k_T^* \in \arg \min_{1 \le k \le min\{m, T-1\}} \left(A + R_k^{\mathrm{bag}}\right) \frac{T-1}{T-k}.$$

Suppose the marginal regularity condition holds, $A > 0$ is fixed, and $R_k^{\mathrm{bag}}$ is uniformly bounded. If the pool size is large enough that the relevant range satisfies $T^{1/2} \ll m$, then

$$k_T^* = \Omega(T^{1/3}), \qquad k_T^* = O(T^{1/2}).$$

Thus

$$T^{1/3} \lesssim k_T^* \lesssim T^{1/2}.$$

If $m$ is fixed or grows more slowly than $T^{1/2}$, the upper rate is truncated by the constraint $k \le m$. □

*Proof.* The loss decreases from $k$ to $k+1$ if and only if

$$L_{k+1}^{\mathrm{bag}} < L_k^{\mathrm{bag}}.$$

Using

$$L_k^{\text{bag}} = \left(A + R_k^{\text{bag}}\right) \frac{T-1}{T-k},$$

cross-multiplication gives the equivalent condition

$$(T-k)\Delta R_k > A + R_k^{\text{bag}}.$$

For the lower rate bound, the marginal lower envelope implies

$$(T-k)\Delta R_k \gtrsim Tk^{-3}$$

for $k \ll T$. Since $A + R_k^{\text{bag}}$ is uniformly bounded above, the inequality still holds for all $k \leq cT^{1/3}$ for some constant $c > 0$. Hence the loss is decreasing up to the cube-root scale, so the minimizer cannot occur below it:

$$k_T^* = \Omega(T^{1/3}).$$

For the upper rate bound, the marginal upper envelope implies

$$(T-k)\Delta R_k \lesssim Tk^{-2}.$$

Since $A + R_k^{\text{bag}} \geq A > 0$, this quantity is smaller than $A + R_k^{\text{bag}}$ for all $k \geq CT^{1/2}$, for some sufficiently large constant $C$. Hence the loss is increasing beyond the square-root scale, so

$$k_T^* = O(T^{1/2}).$$

□

# H Additional Discussion of Monte Carlo Simulations Results

Here, all the plots for both relative RMSFE as well as $Q^{RSM}(k)$ across the scenarios are presented.

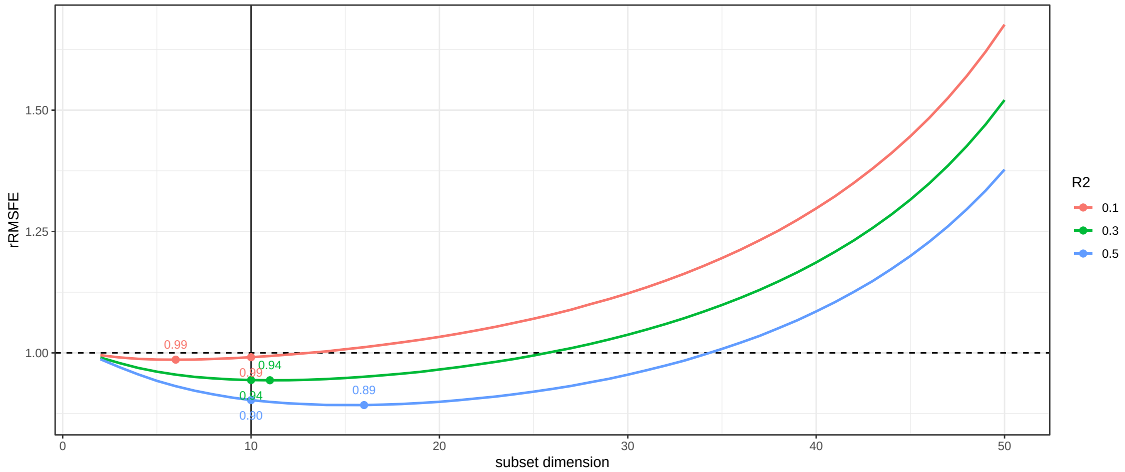


Figure 16: Relative forecasting performance ($T = 100$, $\rho = 0$, $\phi_f = 0.3$, $\phi_\eta = 0.7$)

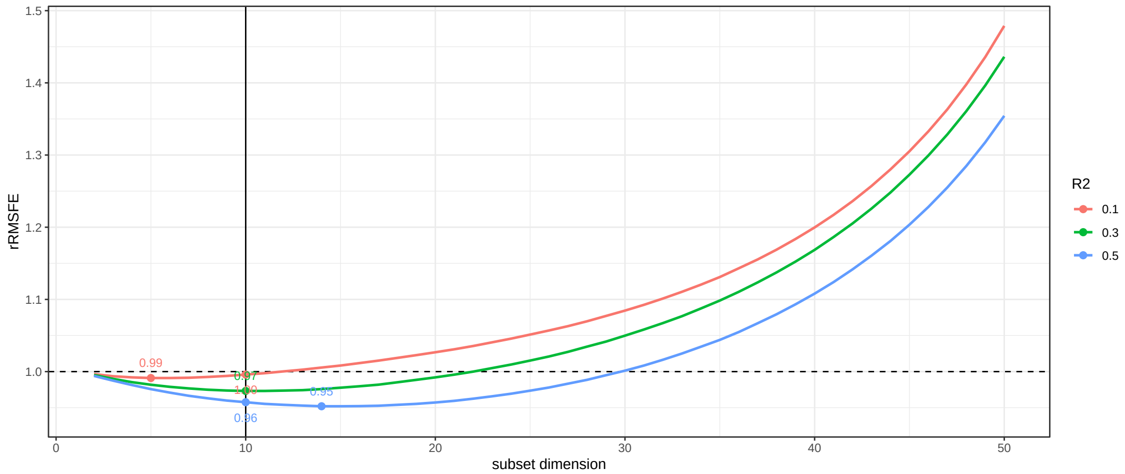


Figure 17: Relative forecasting performance ($T = 100$, $\rho = 0$, $\phi_f = 0.7$, $\phi_\eta = 0.3$)

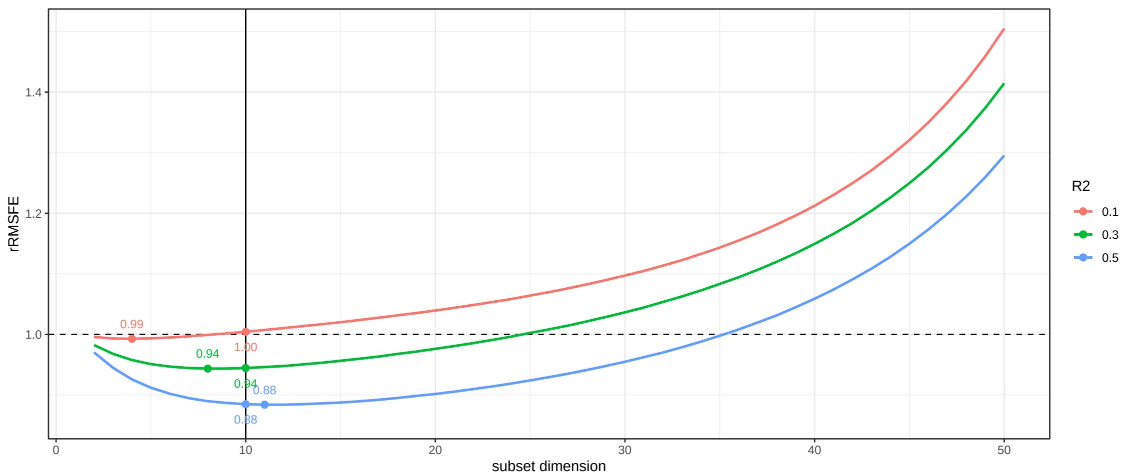


Figure 18: Relative forecasting performance ($T = 100$, $\rho = 0.8$, $\phi_f = 0.7$, $\phi_\eta = 0.3$)

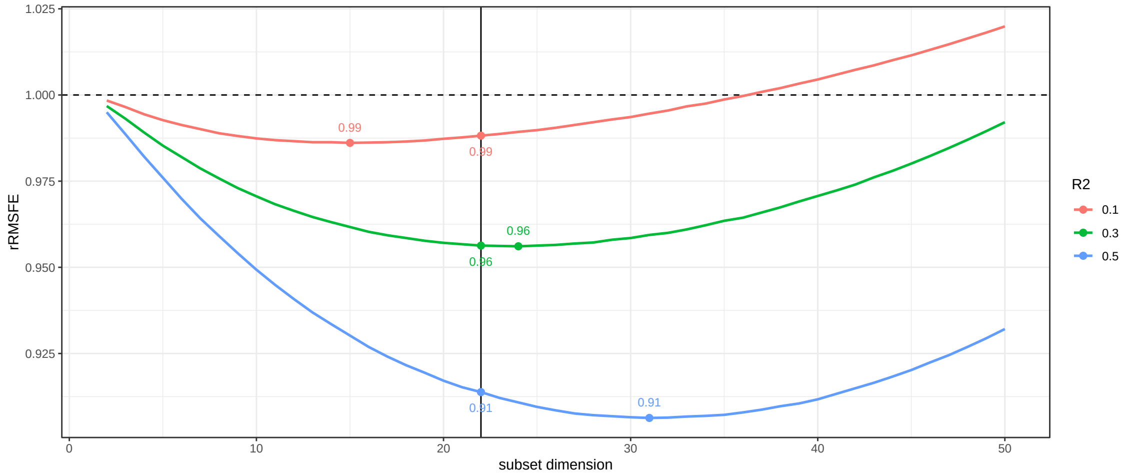


Figure 19: Relative forecasting performance ($T = 500$, $\rho = 0$, $\phi_f = 0.3$, $\phi_\eta = 0.7$)

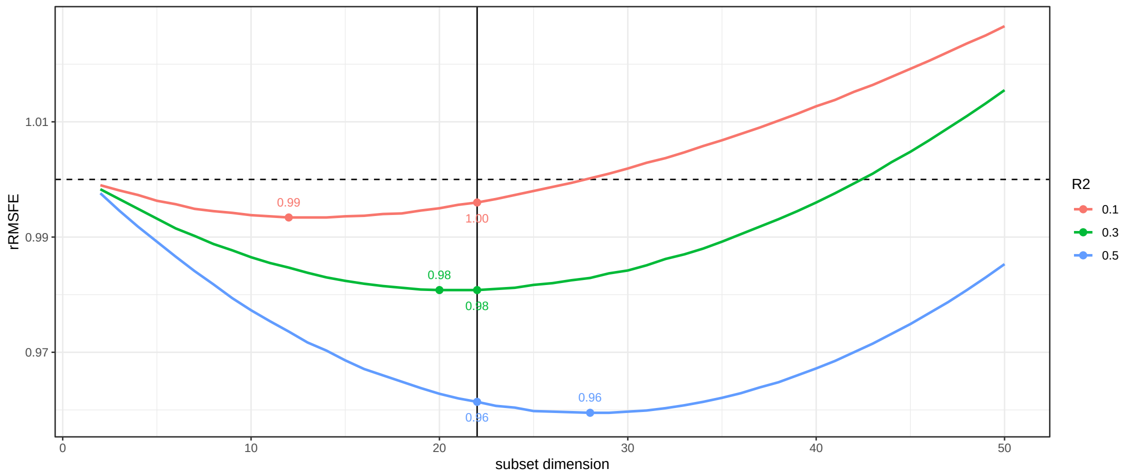


Figure 20: Relative forecasting performance ($T = 500$, $\rho = 0$, $\phi_f = 0.7$, $\phi_\eta = 0.3$)

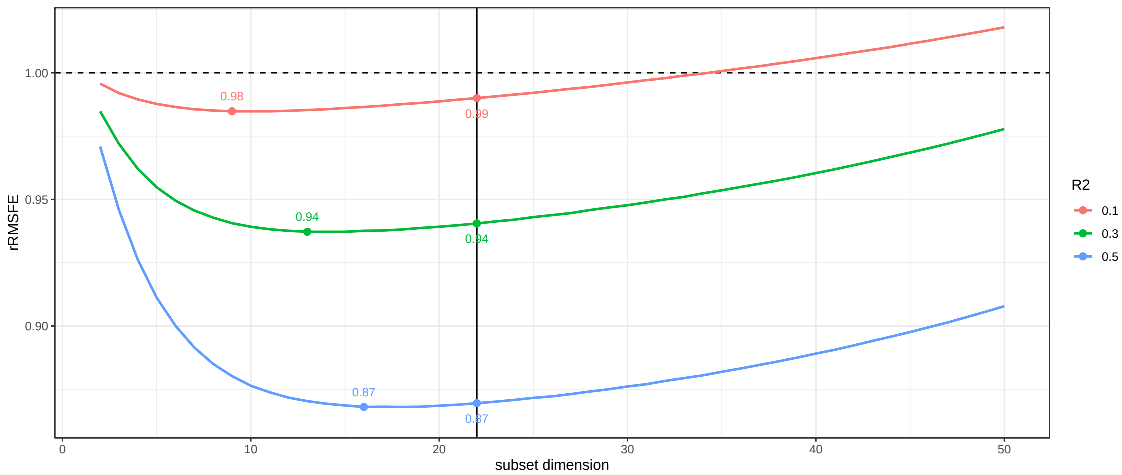


Figure 21: Relative forecasting performance ($T = 500$, $\rho = 0.8$, $\phi_f = 0.7$, $\phi_\eta = 0.3$)

### H.1 $Q^{RSM}(k)$ for Different Scenarios

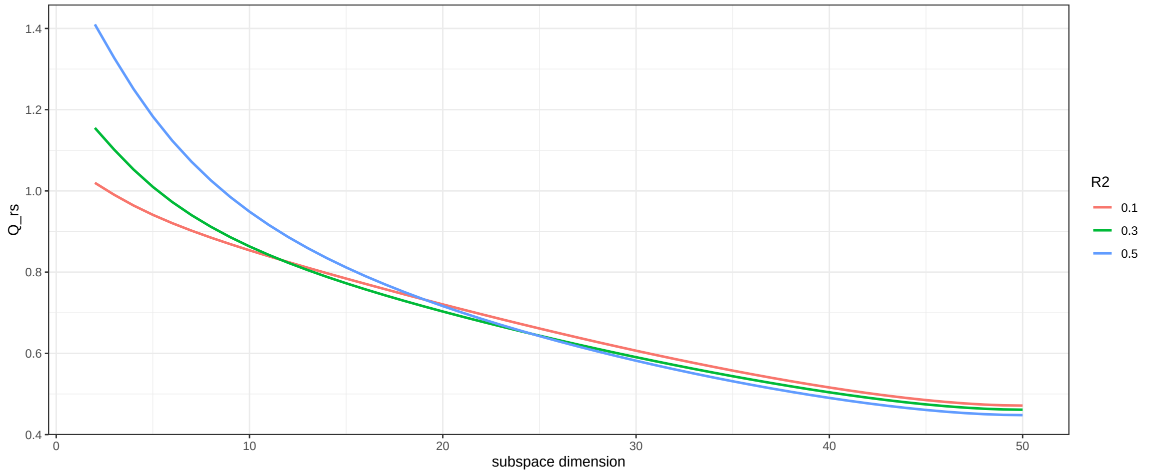


Figure 22: $Q^{RSM}$ $(T = 100,\ \rho = 0,\ \phi_f = 0.3,\ \phi_\eta = 0.7)$

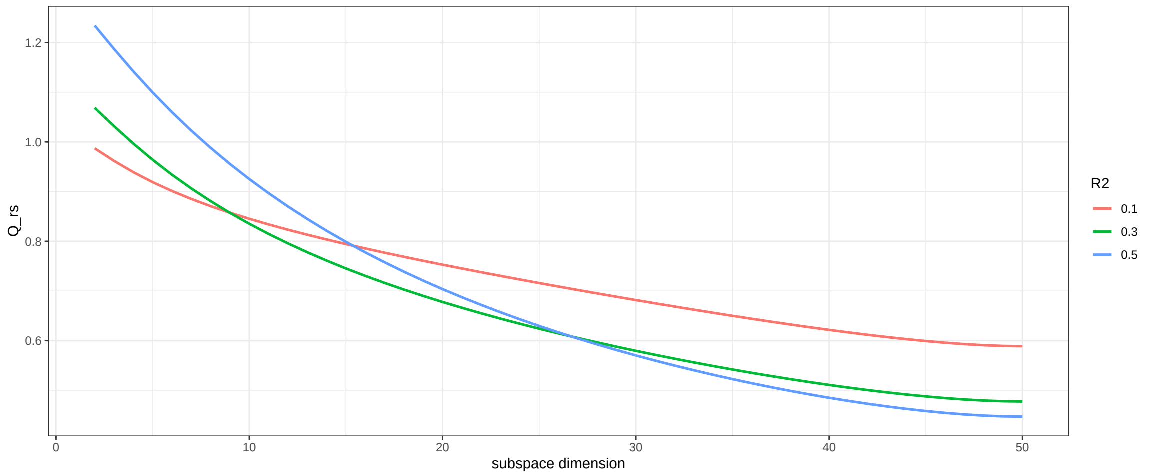


Figure 23: $Q^{RSM}$ $(T = 100,\ \rho = 0,\ \phi_f = 0.7,\ \phi_\eta = 0.3)$

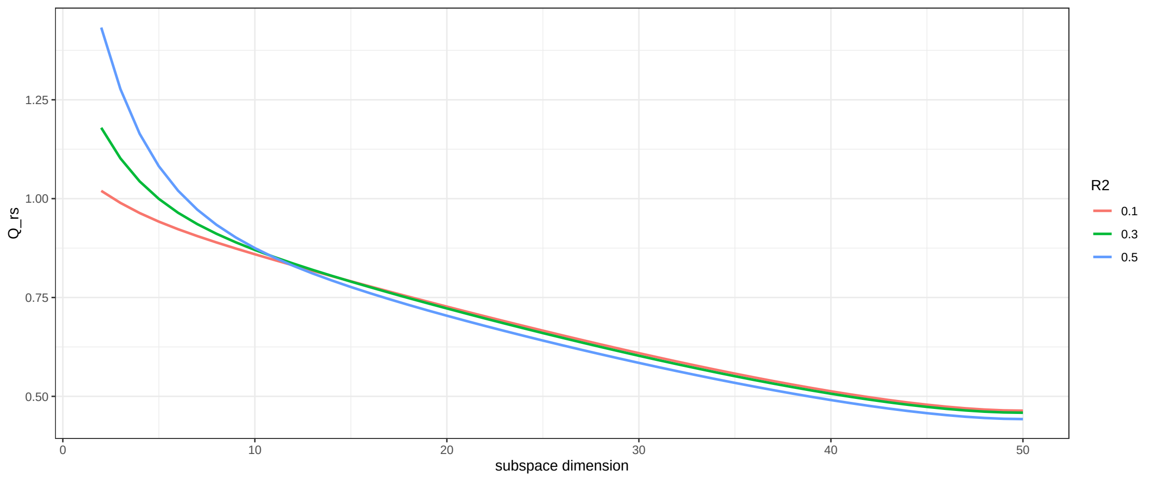


Figure 24: $Q^{RSM}$ $(T = 100,\ \rho = 0.8,\ \phi_f = 0.3,\ \phi_\eta = 0.7)$

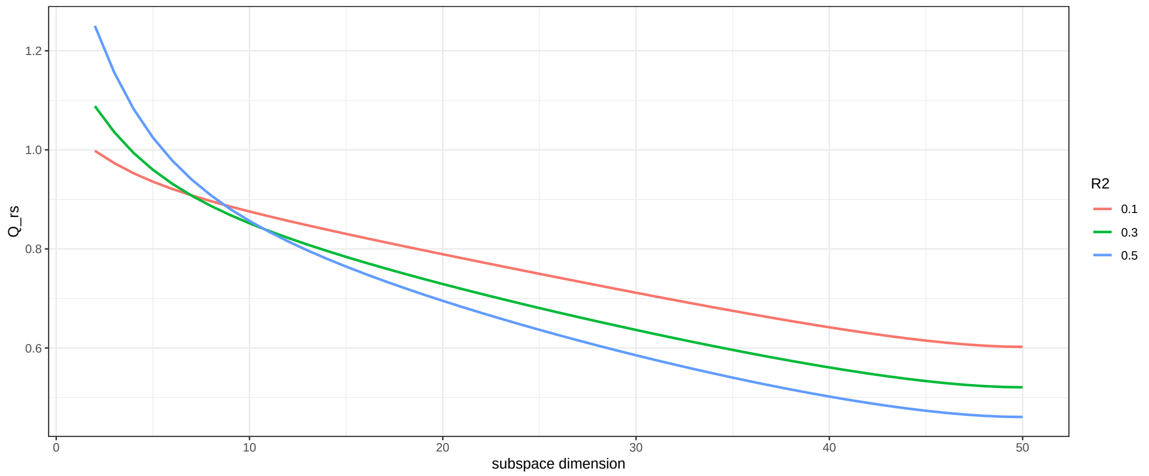


Figure 25: $Q^{RSM}$ ($T = 100$, $\rho = 0.8$, $\phi_f = 0.7$, $\phi_\eta = 0.3$)

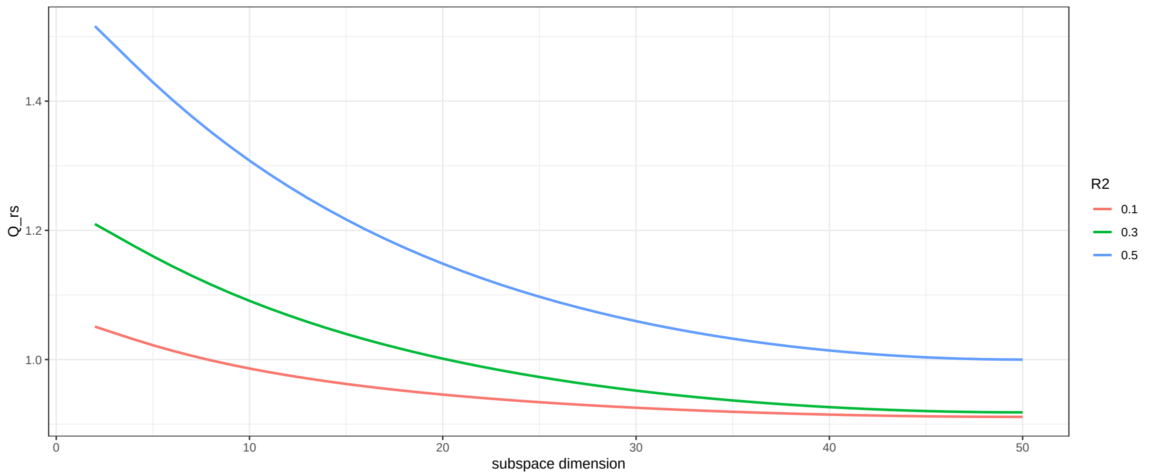


Figure 26: $Q^{RSM}$ ($T = 500$, $\rho = 0$, $\phi_f = 0.3$, $\phi_\eta = 0.7$)

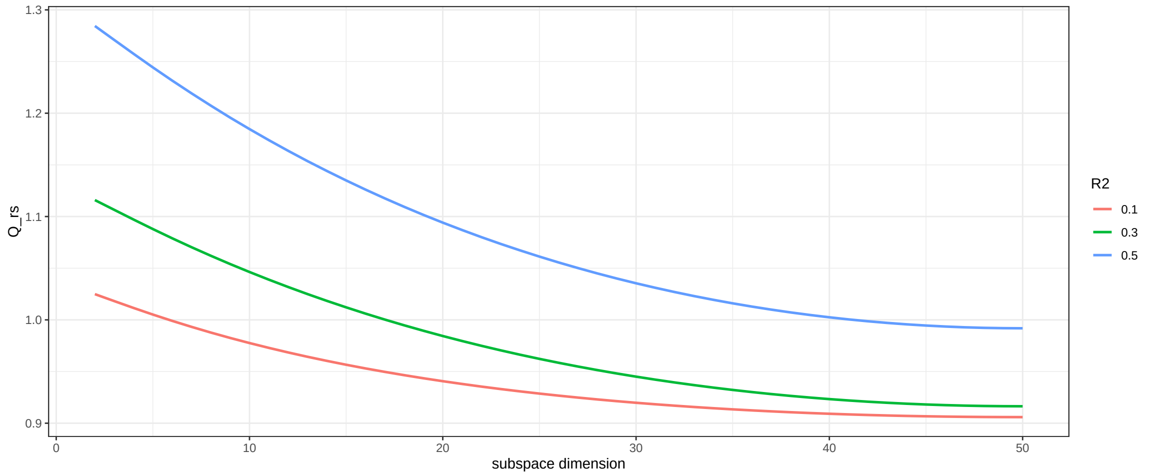


Figure 27: $Q^{RSM}$ ($T = 500$, $\rho = 0$, $\phi_f = 0.7$, $\phi_\eta = 0.3$)

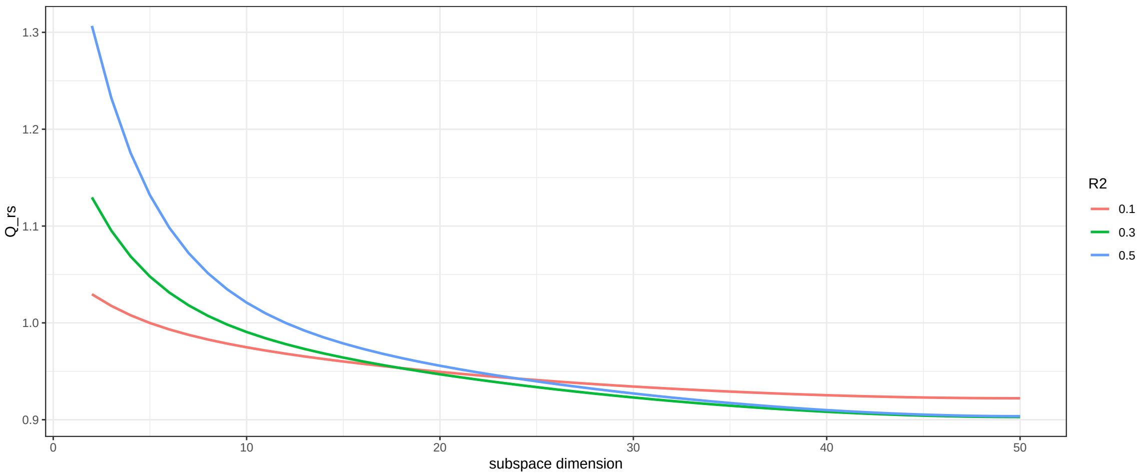


Figure 28: $Q^{RSM}$ ($T = 500$, $\rho = 0.8$, $\phi_f = 0.7$, $\phi_\eta = 0.3$)

# I Robustness Checks

### I.0.1 RMSFE for Different Subset Dimensions $k$

In Figures 29 and 30, the RMSFE metrics for each subset $k \in [2, 55]$ are depicted with or without the COVID-19 quarters. Similar to Monte Carlo evidence, both curves have a U-shape, further illustrating that the optimal $k^*$ lies somewhere in between. The exception is for $k = 37$, where a sudden jump is observed, which can be treated as an outlier. In our baseline scenario, we choose $k^*$ as the nearest integer to the $\sqrt{T}$, which corresponds to $k^* = 9, 10, 11$. Again, the imposed rule-of-thumb to select $k^*$ is in line with the empirical findings in Boot and Nibbering (2019), that $k^*$ is somewhat stable over time.

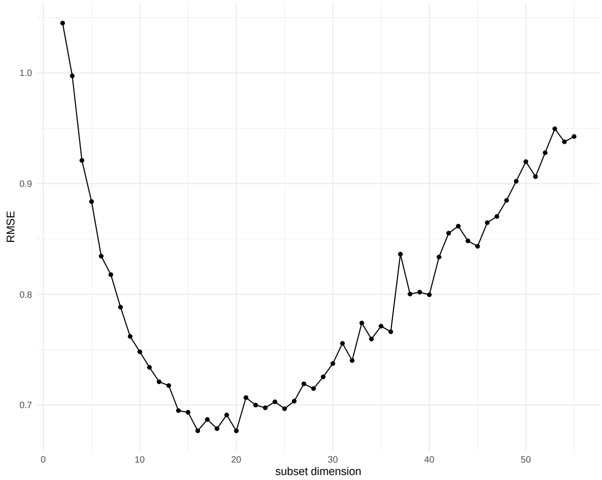


Figure 29: RMSFE full data

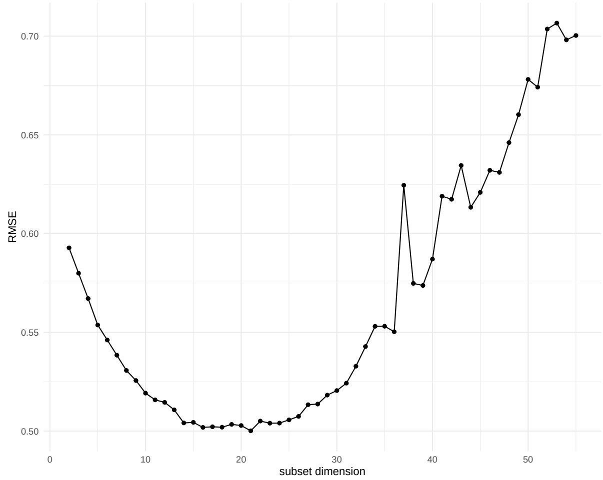


Figure 30: RMSFE without COVID-19

As can be seen in Table 1 and Table 2, the rule-of-thumb rule produces the RMSFE of 0.74 for the full sample and 0.53 for the COVID-19 restricted sample[8]. The corresponding RMSE minima are 0.68 for $k = 18$ and 0.5 for $\forall k \in [14, 26]$, respectively. In both cases, the optimal subset, which is characterized by the lower RMSFE, is higher that the ad-hoc $k^*$ selection. Even though, the proposed $k^*$ choice during the COVID-19 pandemic is able to produce more or less accurate results compared to the mean, the bigger subset dimension can improve the forecasting performance even further.

These findings show that the grid search over all possible $k$, implemented around $k = \sqrt{T}$, can improve forecasting performance, although sometimes only marginally (if COVID-19 is excluded).

[8]Recall, that RSM randomly selects variables. When implementing this exercise, a different seed is used, yielding 0.73 and 0.51, which are even closer to the corresponding minimum values

# J Global and Conditional Weights

For illustrative purposes, we show the correspondence between the global weight $\bar{w}_j^{zb}$ and conditional weight $m_j$ using All Employees: Retail Trade variable (FRED-md mnemonics USTRADE). This variable is chosen (without loss of generality) to depict the difference in the evolution of the global weight compared to the equal weighting.

Recall that both weights are interconnected via the following expression (Equation 41 in the main text):

$$\bar{w}_j^{\text{zb}} = \frac{k}{N} m_j \quad \Longrightarrow \quad \boxed{\bar{w}_j^{\text{zb}} > \tfrac{1}{N} \iff m_j > \tfrac{1}{k}}$$

Therefore, when comparing to the mean, one can use both weights (global and conditional), compared to different values (the equal weight $\frac{1}{N}$ and the subset equal weight $\frac{1}{k}$, respectively). Both of these comparisons should provide the same results, and the plots should look similar to each other. To be more specific, the global weight should be scaled to $\frac{k}{N}$, which in our case is approximately equal to 0.1, since $k \in [9, 11]$ and $N \approx 120$ for each vintage.

In Figure 31 and Figure 32, the time-varying USTRADE weights are presented.

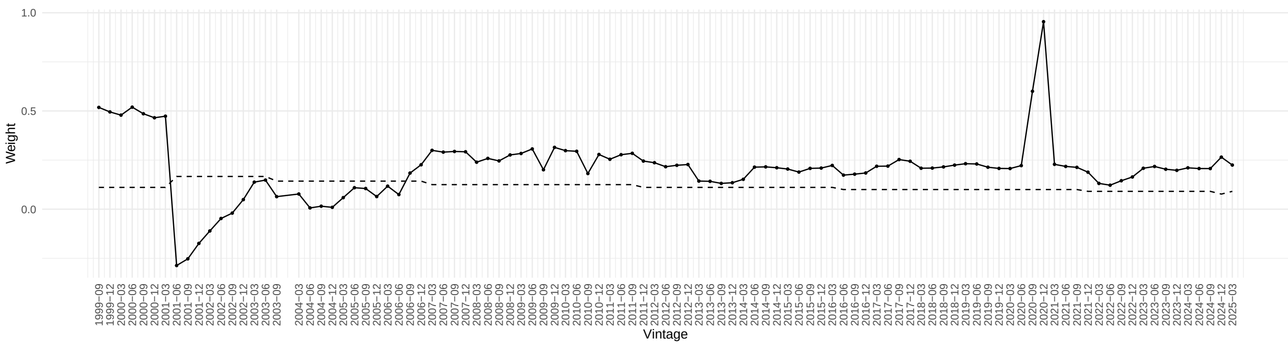


Figure 31: USTRADE: Global Weight against Equal Weight

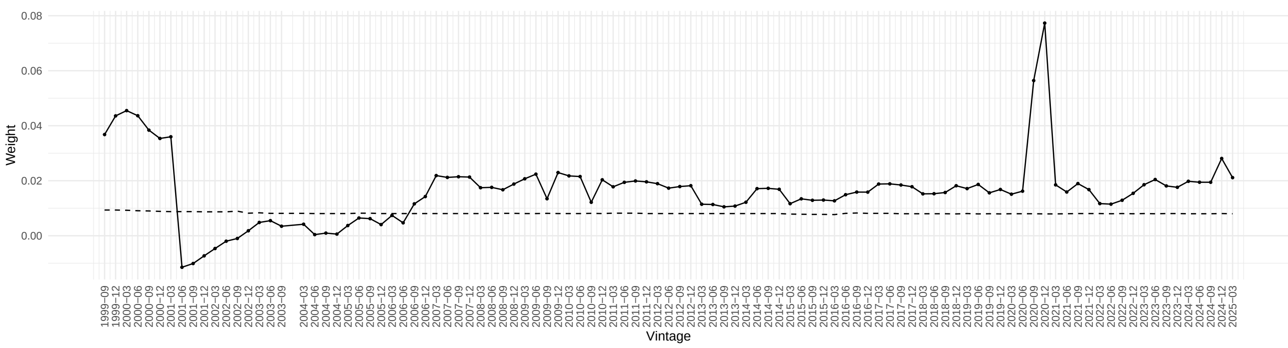


Figure 32: USTRADE: Conditional Weight against subset Equal Weight

First, as expected, the plots show the same dynamics. That is for all $\bar{w}_j^{zb} > \frac{1}{N}$ we can observe that $m_j > \frac{1}{k}$. Moreover, in our case, both $N$ and $k$ are somewhat stable over time. That happens because the number of time series in the FRED-md does not vary substantially with rare occasions of dropouts. Whereas the established $k^*$ changes at the speed $\sqrt(T)$, which is also slow. Thus, both dotted lines in Figure 31 and 32 appear to be straight, even though some slight jumps occurred. If we choose $k^*$ differently (say via cross-validation) then the subset dimension can significantly vary over time (the dotted line in Figure 32, but the established correspondence between $\bar{w}_j^{zb}$ and $m_j$ would still hold.